Creative Ownership in the Age of AI

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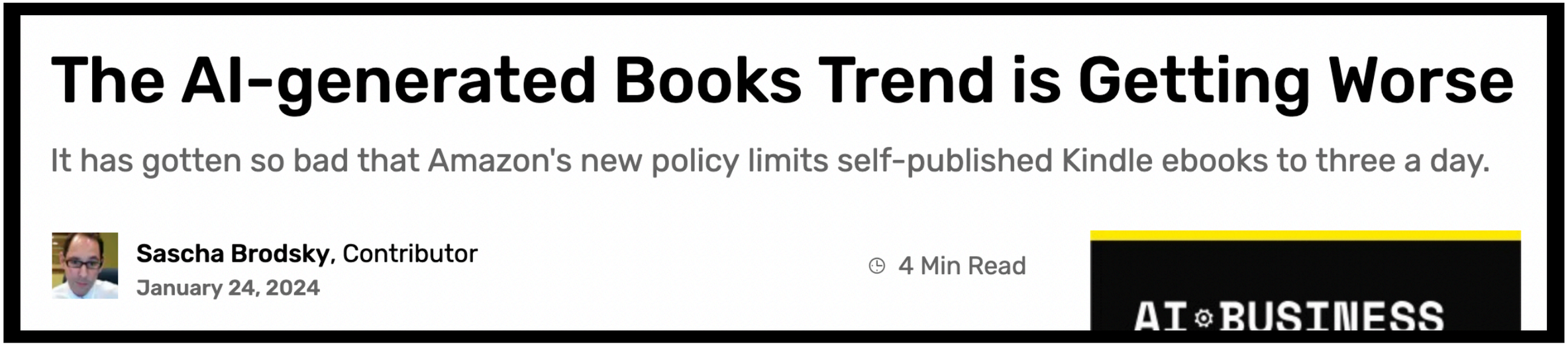
Introduction

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- Novels



The AI-generated Books Trend is Getting Worse

It has gotten so bad that Amazon's new policy limits self-published Kindle ebooks to three a day.

Sascha Brodsky, Contributor
January 24, 2024

4 Min Read

AI BUSINESS

Introduction

Generative AI systems—trained on large corpora of human-authored works—are increasingly used to produce new creative content.

- Novels
- Cartoons

Generating New Yorker-Style Cartoons with AI

Harnessing the power of OpenAI's ChatGPT to Transform Cartoon ideas into Reality



Georgiy Ermakov Follow 5 min read · Oct 25, 2023

Introduction

Generative AI systems—trained on large corpora of human-authored works—are increasingly used to produce new creative content.

- Novels
- Cartoons
- Even AI "actors"

Tilly Norwood: how scared should we be of the viral AI 'actor'?

A bunch of code is being pushed as the next Scarlett Johansson, a creation that is already causing pushback from real human actors

Tilly Norwood



A 2025 AI-generated image of Tilly Norwood, taken from the character's Instagram account

Introduction

Generative AI systems—trained on large corpora of human-authored works—are increasingly used to produce new creative content.

- Novels
- Cartoons
- Even AI "actors"

The creators of those original works are pushing back.

- Andersen v. Stability AI
- Disney Enterprises v. Midjourney
- UMG Recordings, Inc. v. Suno, Inc.

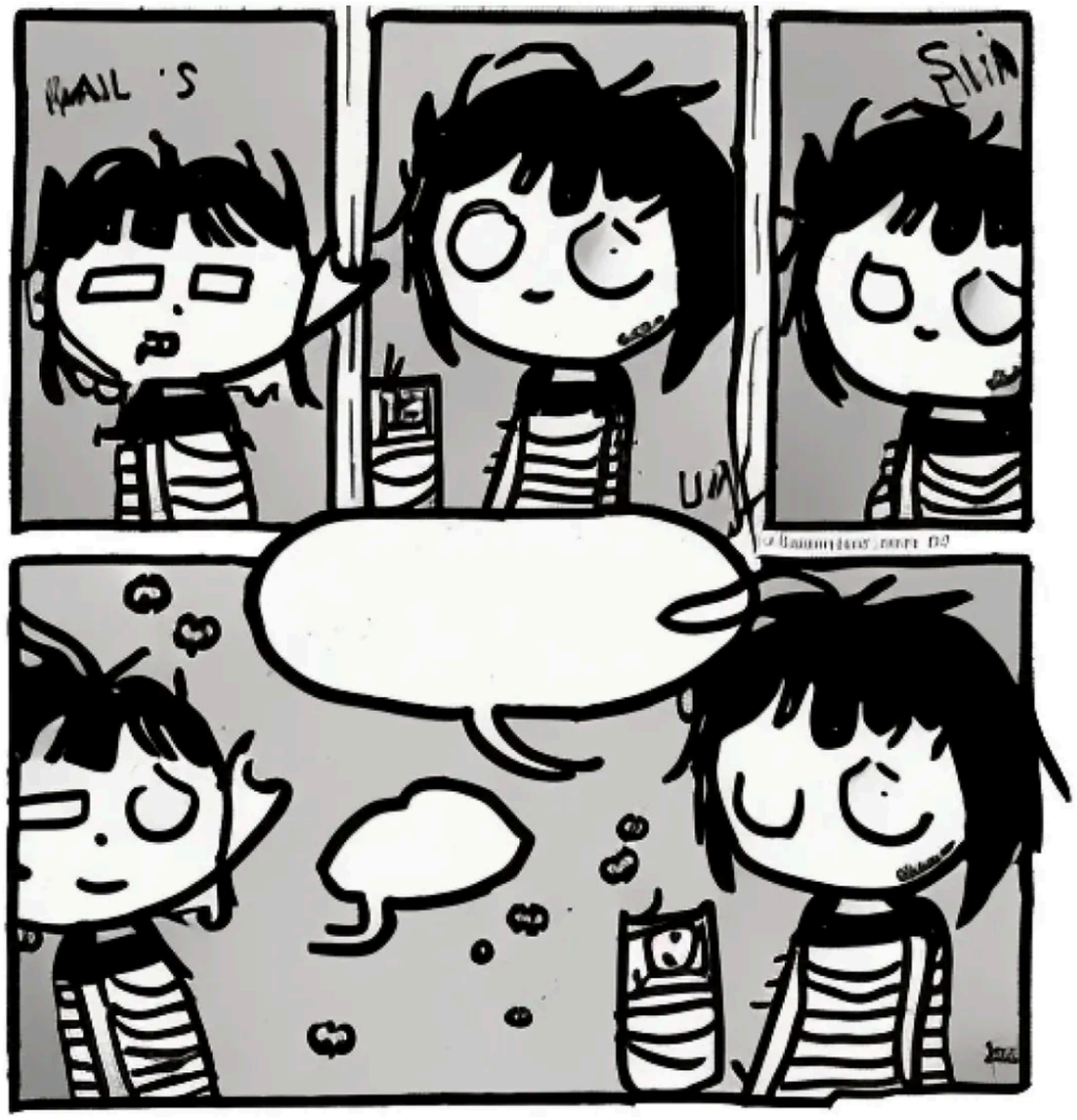
Andersen v. Stability AI

A recent example of my work



Sarah Andersen, *NYTimes*, 2022

An A.I. generated image that I created when I used my name as a prompt



It's not perfect — but it has captured the signature elements of my drawing style. The black bangs, striped shirt and wide eyes are immediately recognizable. As A.I. gets increasingly sophisticated, I am growing more concerned about what's possible.

Math and AI

"Over the course of generations, [research mathematics] has built a **large corpus of sophisticated ideas, methods, abstractions**, and other tools...In turn, modern technologies and sciences are based on mathematical tools...

...Often these solutions are announced in a rush, leaving no time for a proper writeup, the isolation of new methods and ideas, and citing relevant previous work of others. **As in all creative professions, this raises severe attribution and plagiarism questions**...

We are witnessing a **general threat to intellectual work**...The issues the mathematical community faces now are similar to issues that other scientific and creative professions are facing..."

25 Fields Medalists, *Le Monde*, 2026

Artur **Avila** (Fields Medal 2014)
Manjul **Bhargava** (Fields Medal 2014)
Caucher **Birkar** (Fields Medal 2018)
Pierre **Deligne** (Fields Medal 1978)
Yu **Deng** (Fields Medal 2026)
Simon **Donaldson** (Fields Medal 1986)
Hugo **Duminil-Copin** (Fields Medal 2022)
Alessio **Figalli** (Fields Medal 2018)
Martin **Hairer** (Fields Medal 2014)
June **Huh** (Fields Medal 2022)
Maxim **Kontsevich** (Fields Medal 1998)
Elon **Lindenstrauss** (Fields Medal 2010)
Pierre-Louis **Lions** (Fields Medal 1994)
James **Maynard** (Fields Medal 2022)
Curtis **McMullen** (Fields Medal 1998)
Shigefumi **Mori** (Fields Medal 1990)
Ngô Bảo Châu (Fields Medal 2010)
Andrei **Okounkov** (Fields Medal 2006)
Peter **Scholze** (Fields Medal 2018)
Stanislav **Smirnov** (Fields Medal 2010)
Terence **Tao** (Fields Medal 2006)
Maryna **Viazovska** (Fields Medal 2022)
Cédric **Villani** (Fields Medal 2010)
Wendelin **Werner** (Fields Medal 2006)
Efim **Zelmanov** (Fields Medal 1994)

Legal Context

Copyright constrains both the inputs and outputs of generative AI.

1. Training on existing works

- Copying copyrighted works for training may require permission, subject to a contextual fair-use analysis.
- Restricting training can protect creators, but broad restrictions can also limit valuable learning and recombination.

2. Generating new works

- Generated outputs may violate copyright law if they are "substantially similar" to an existing work.
- Copyright law has historically drawn a line between protectable **expression** and unprotectable **style** to preserve creative competition and follow-on innovation.

Our starting point: Existing notions of copyright and intellectual property may not be appropriate for a post-generative AI world.

What is missing? What is the potential concern?

- AI creations can closely imitate style without copying content
- They can be generated **quickly**, at **low cost/effort**, **without expertise**, and are thus very **easily scaled**
- These creations are **only possible** because of the broad corpus of human works the AI systems were trained on.
- Those human works may not be receiving **sufficient credit/attribution**.
- This may **de-incentivize** future (valuable) human creation.

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What is missing? What is the potential concern?

Blanket ban on training on copyrighted inputs:

Your work was critical if the AI system trained on it.

Restriction of generation of "substantially similar" outputs:

Your work was critical if the AI output looks like it.

Our proposed alternative:

Your work was critical if the AI system could not have generated this output without training on your work.

Plan for Talk

1. A (reduced-form) model of AI systems
2. Our definition of permissible generation
3. Properties of permissible generation
4. Permissible generation given large training corpora
5. Extension to creative protection for bodies of works

Part I:

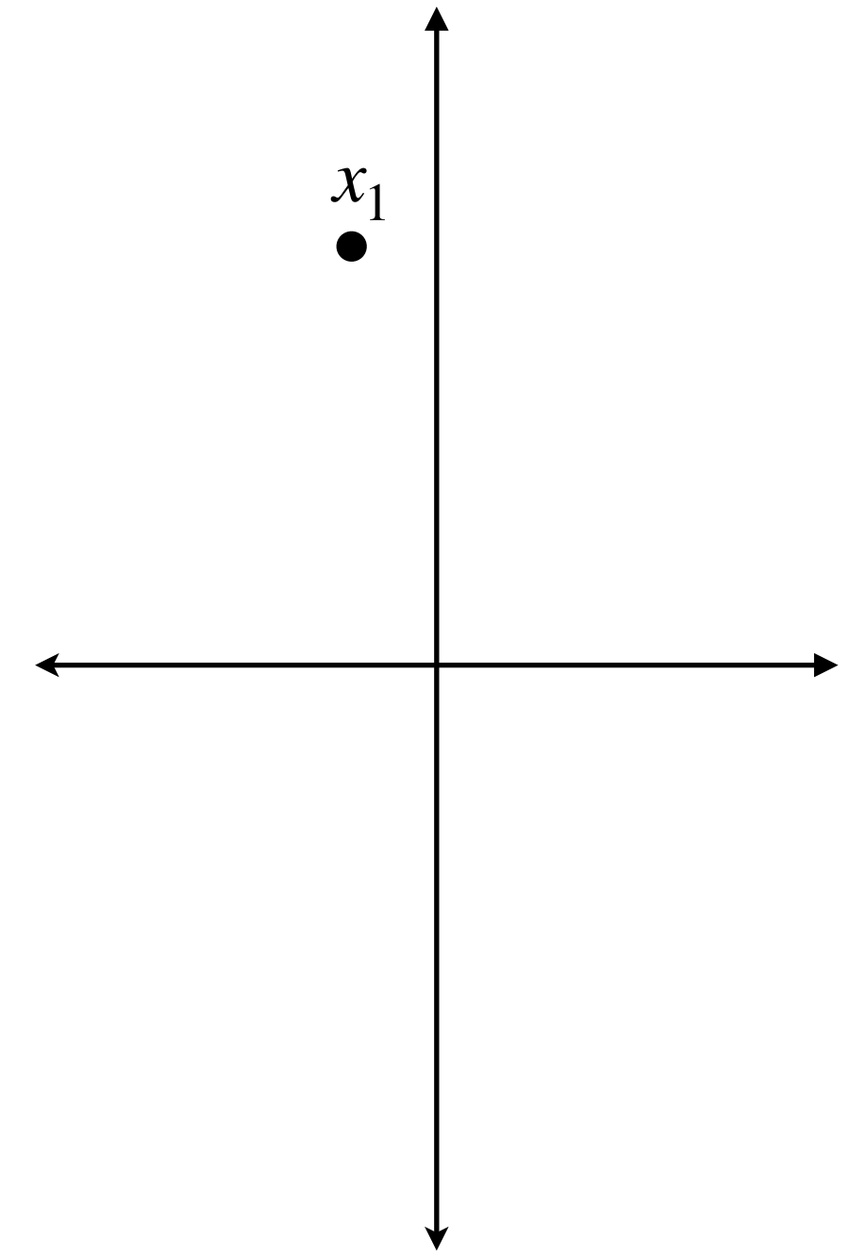
A (reduced-form) model of AI systems

Model

We model AI systems as a technology for generating new creations given an existing collection of works.

Model

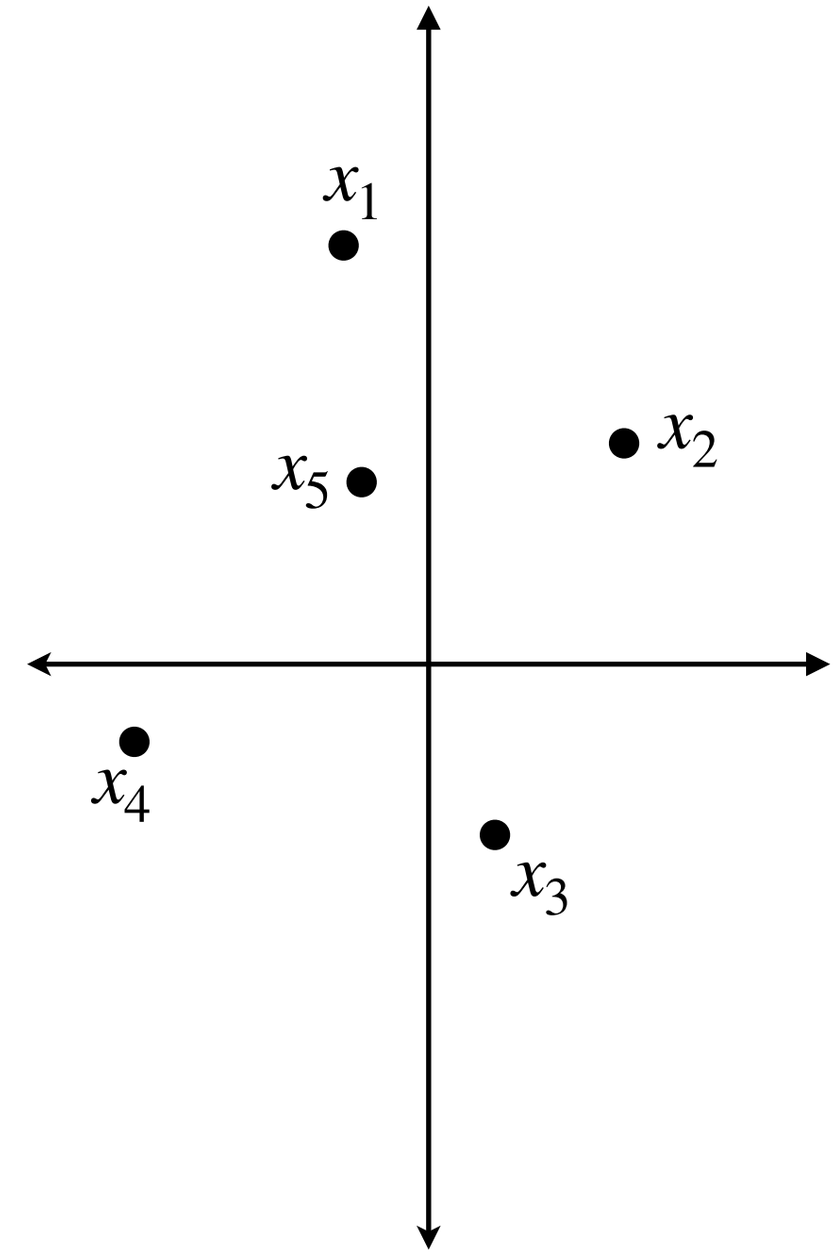
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Model

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- A *corpus* C is a set of works

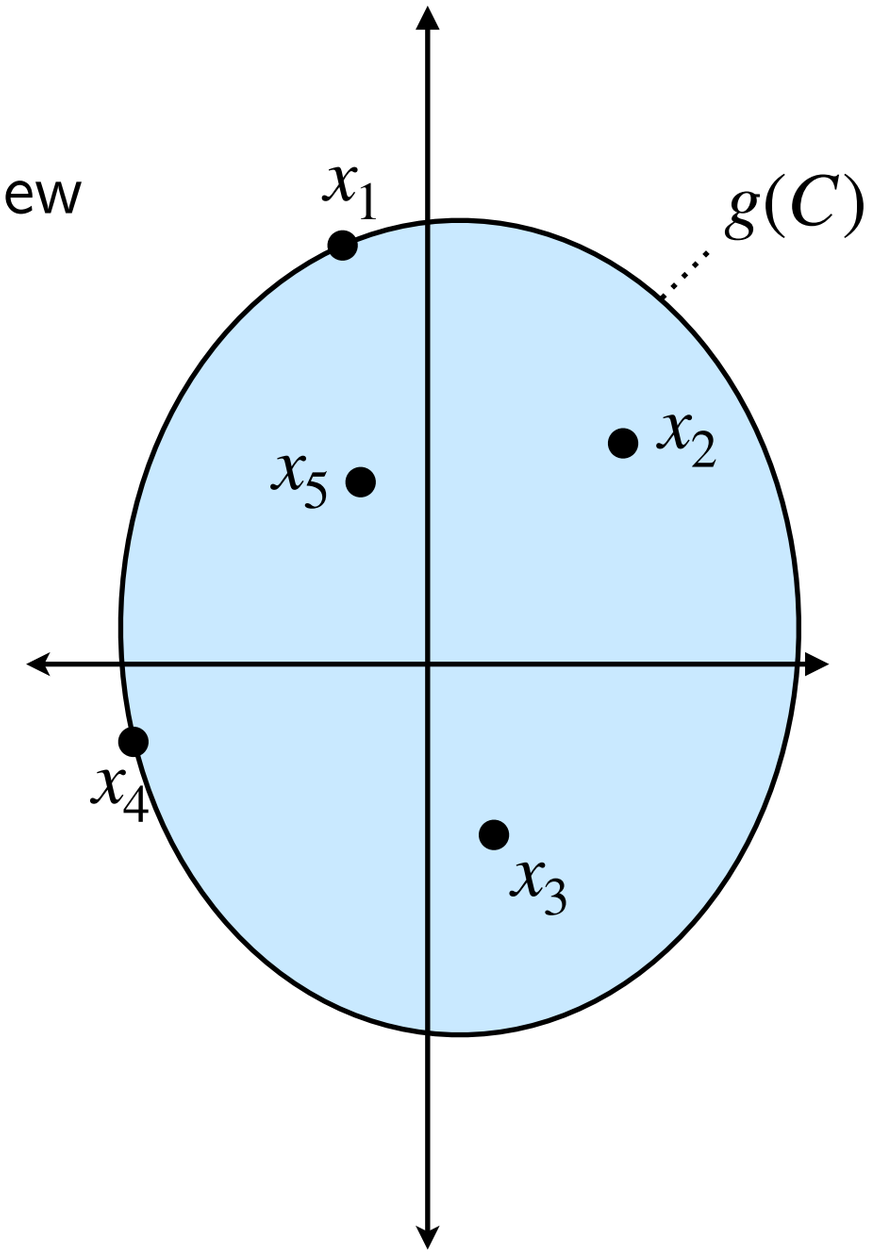
e.g., $C = \{x_1, x_2, x_3, x_4, x_5\}$



Model

- A *work* is a point $x \in \mathbb{R}^d$
- A *corpus* C is a set of works
- A *generator* g is a technology for producing new works from an existing corpus

e.g., $C = \{x_1, x_2, x_3, x_4, x_5\}$



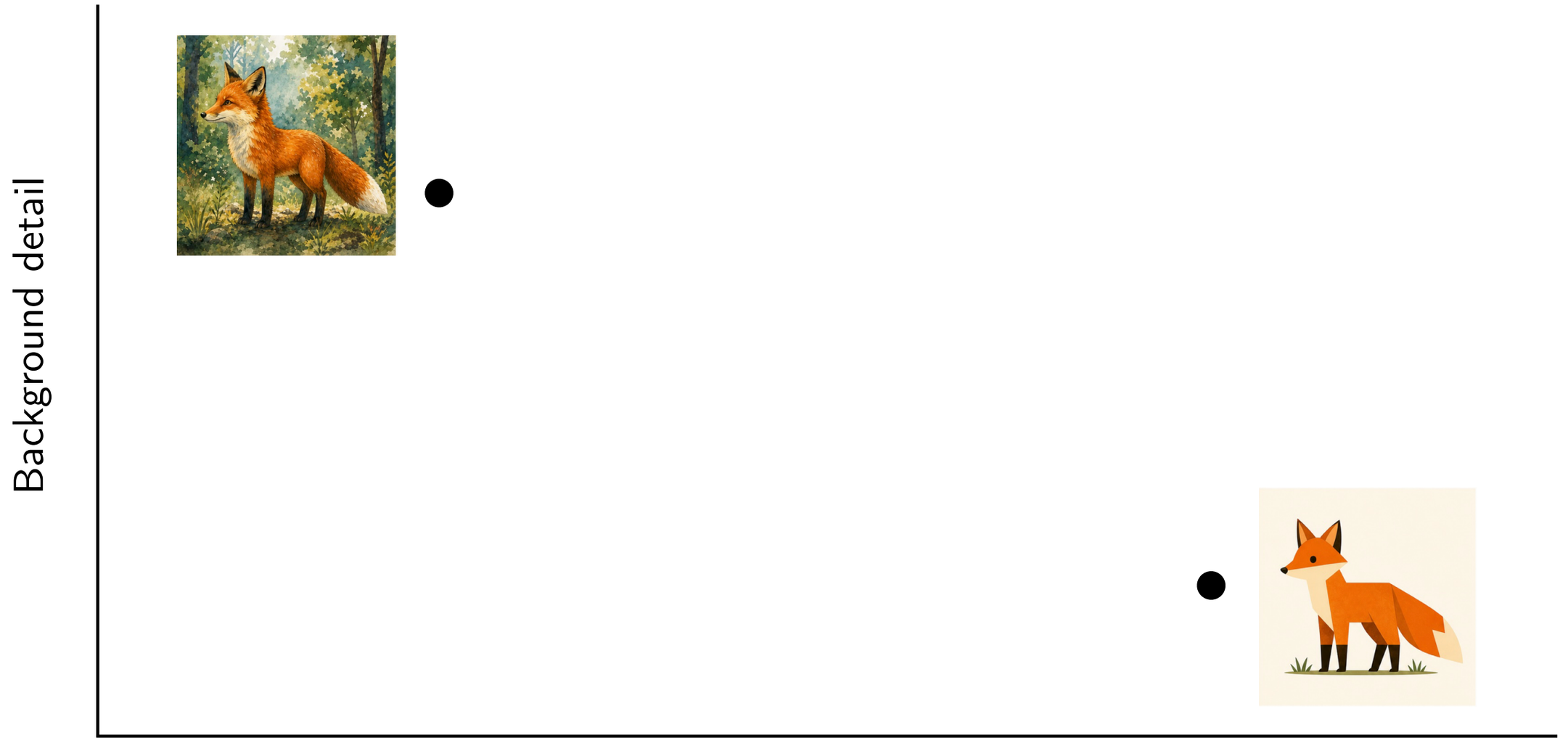
Interpret $g(C)$ as the **range** of possible AI creations for a given training corpus C (not a single run of AI generation)

e.g., "If I train on all books written so far, what are all the books I can generate using this AI system?"

Example Generators

Convex Hull Generator $g_{conv}(C)$

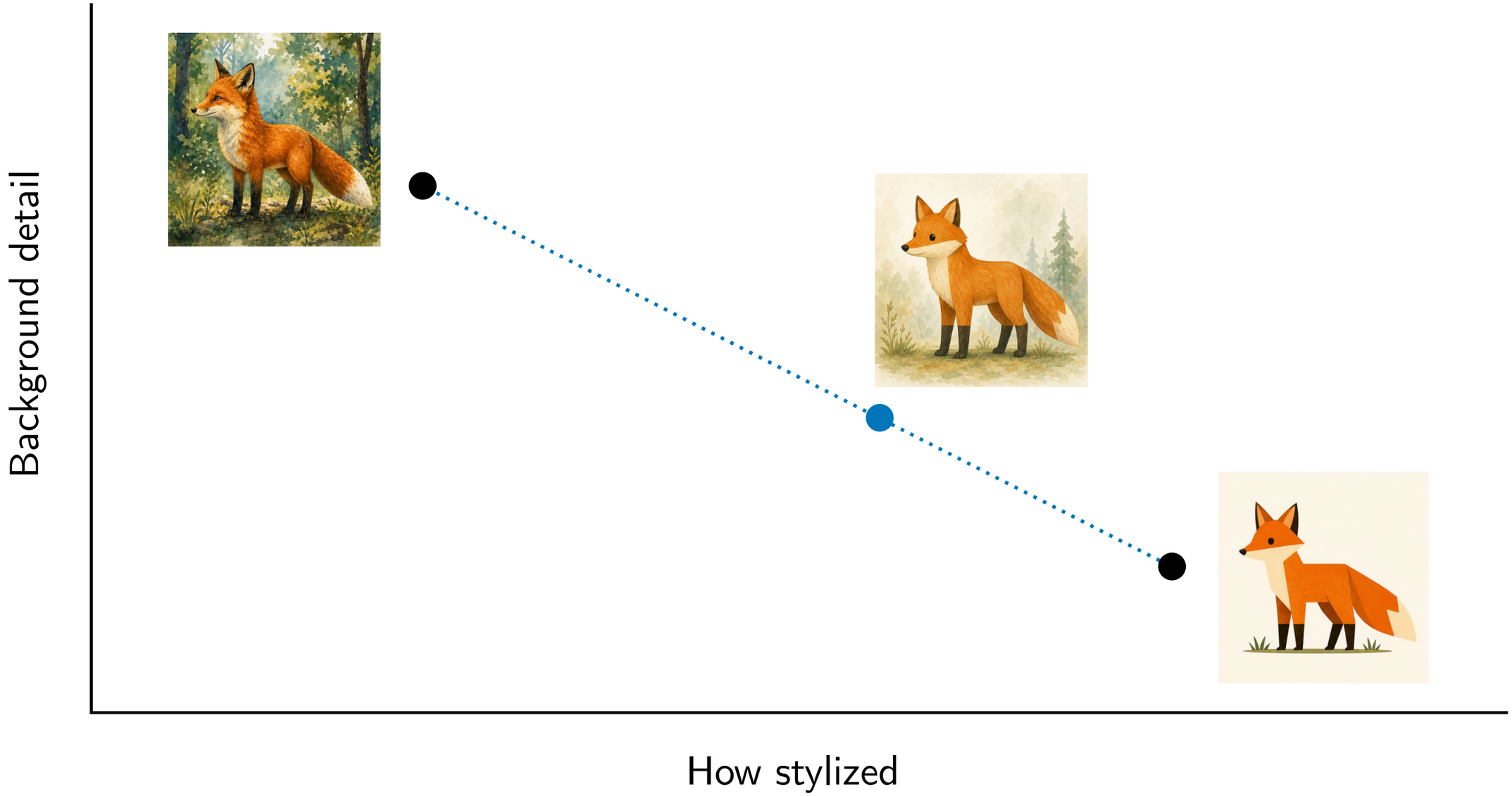
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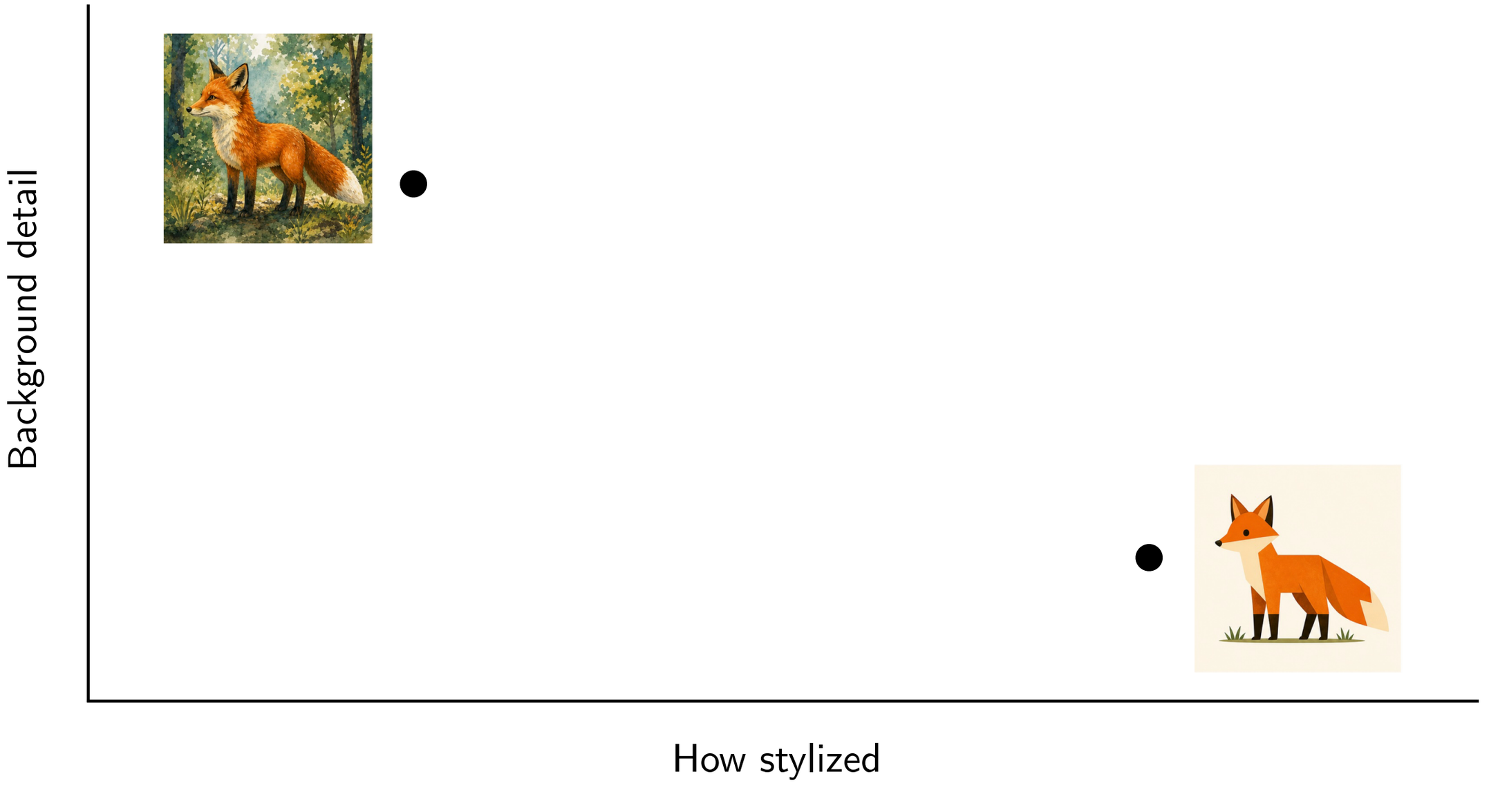
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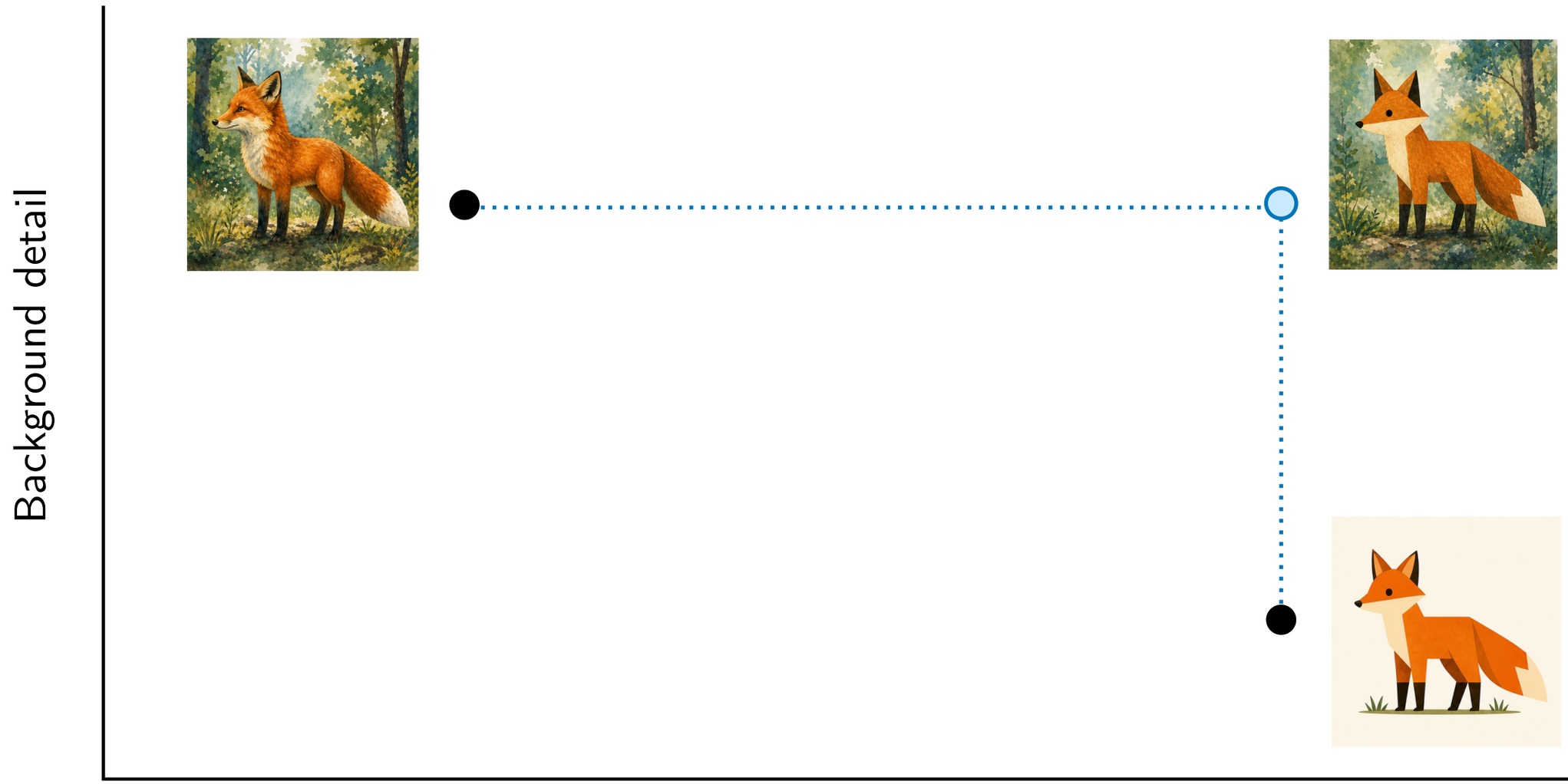
- Recombines features of existing creations



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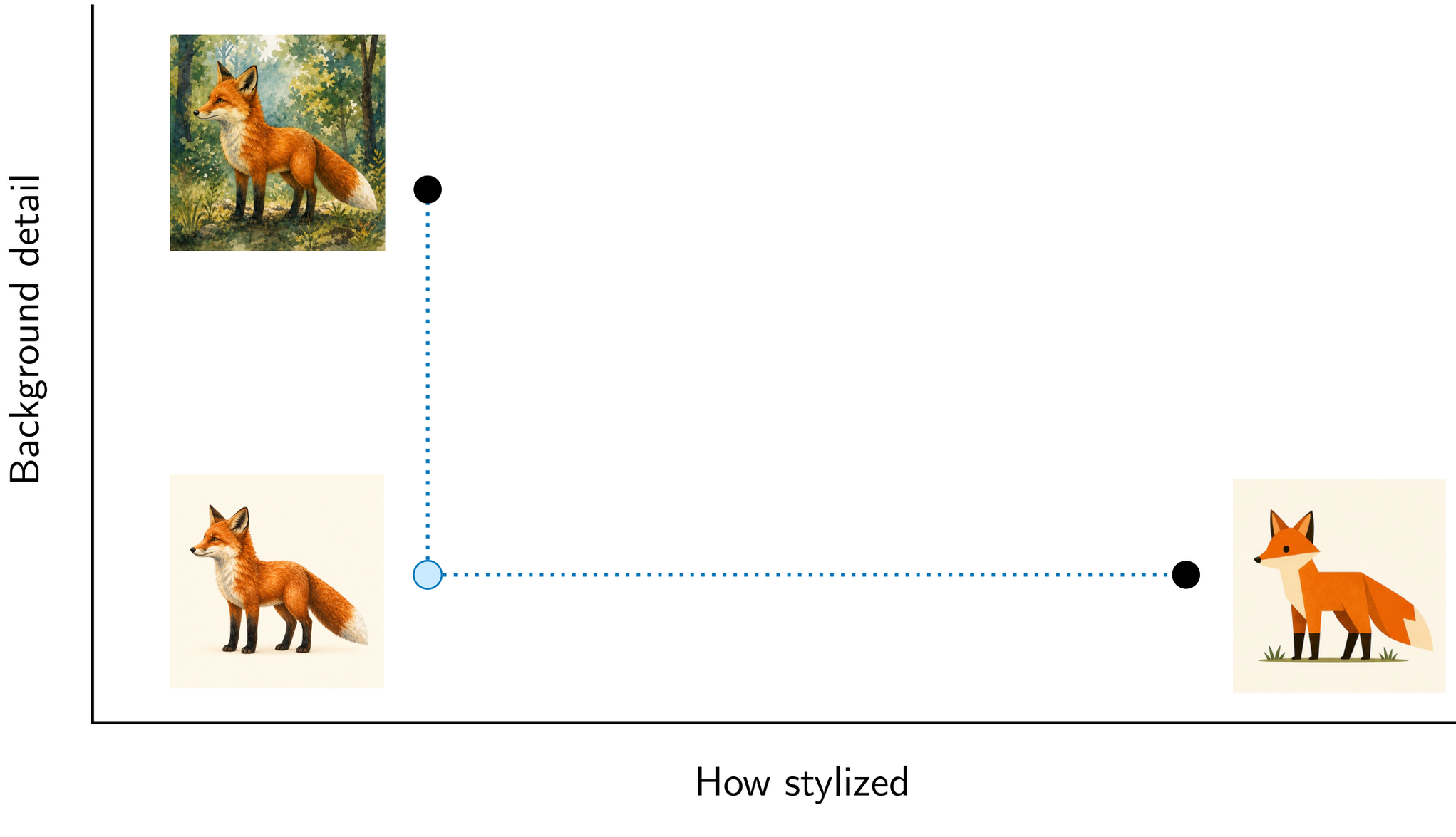
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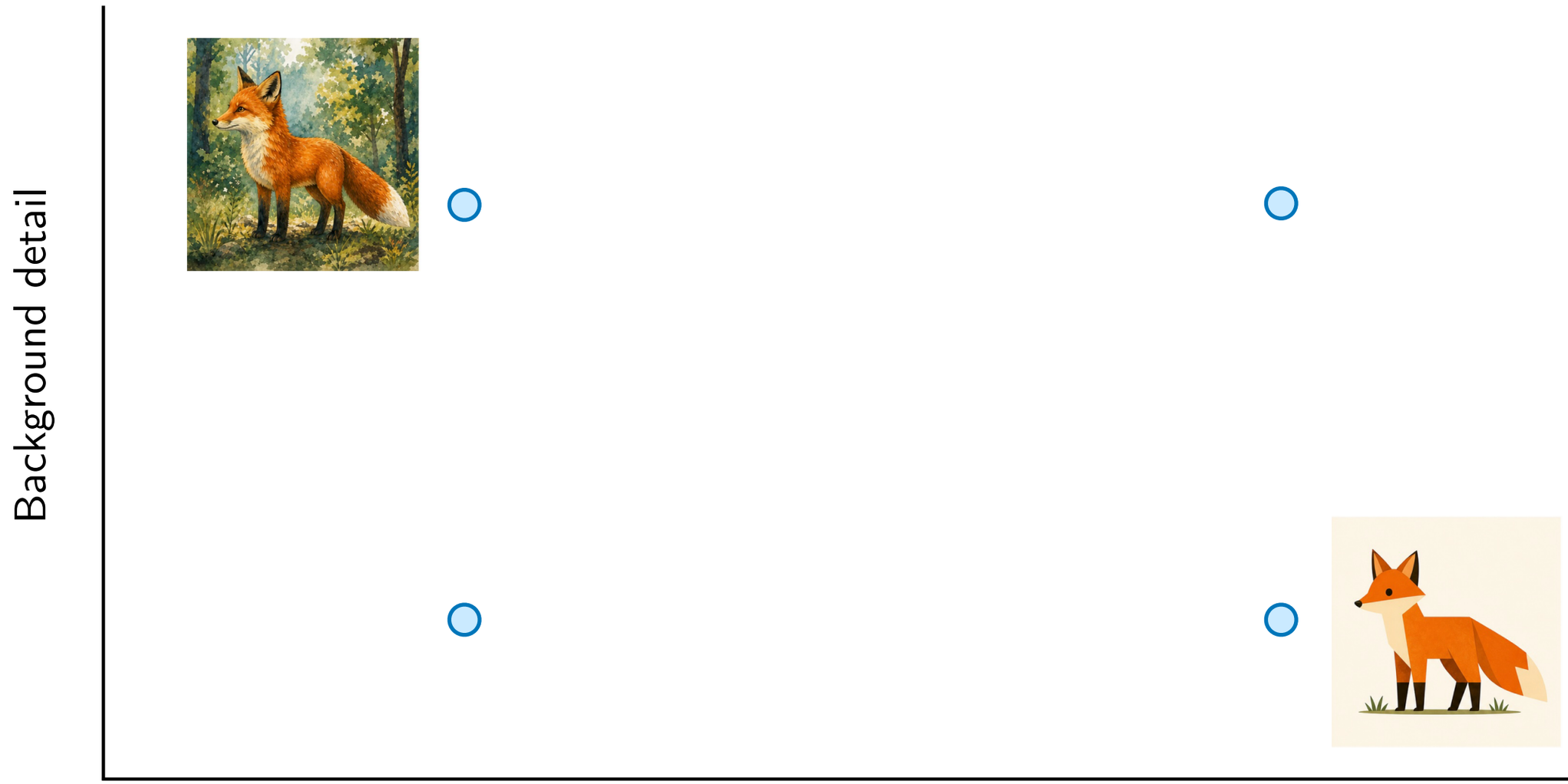
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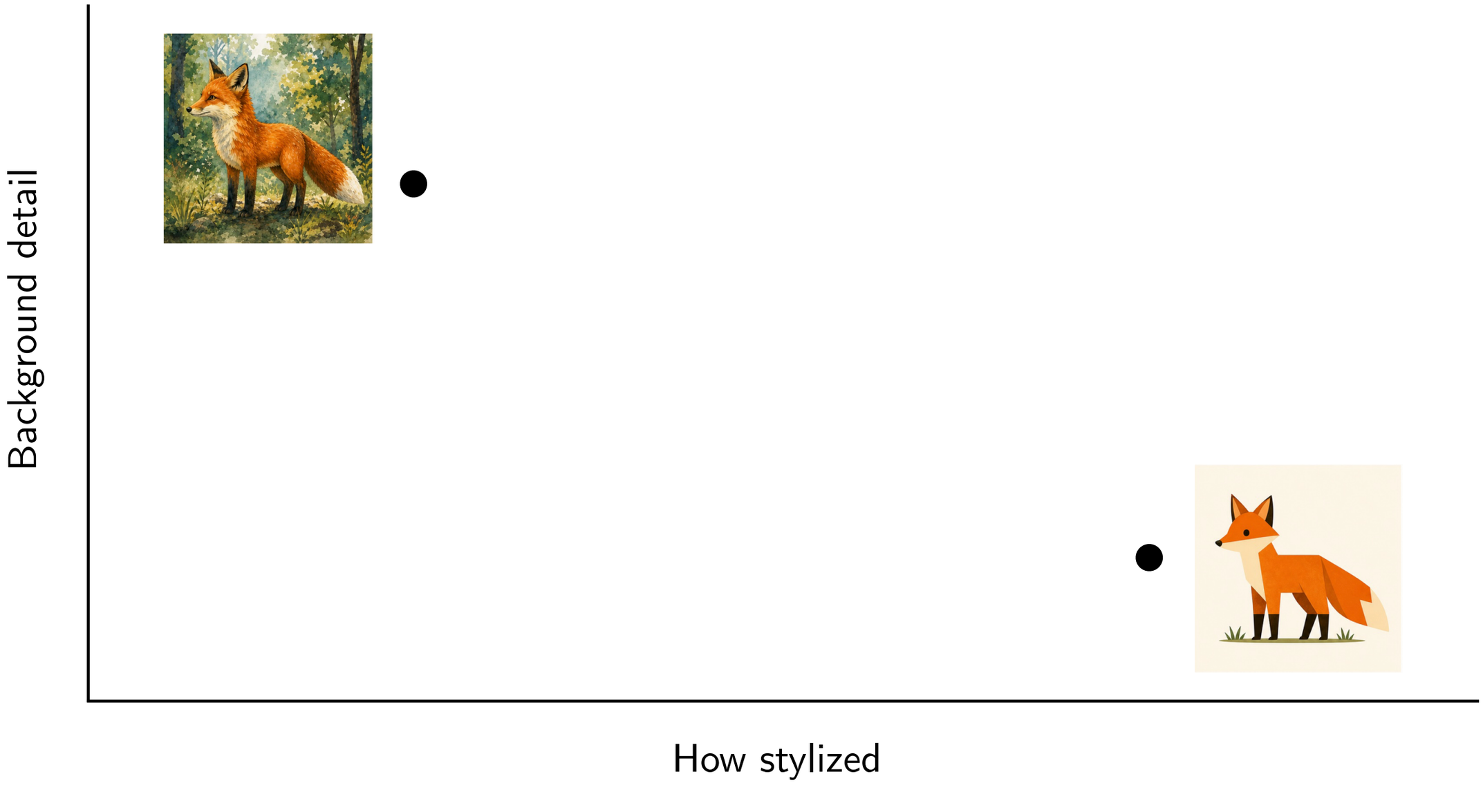
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Example Generators

Box Generator $g_{box}(C) = g_{conv}(g_{splice}(C))$

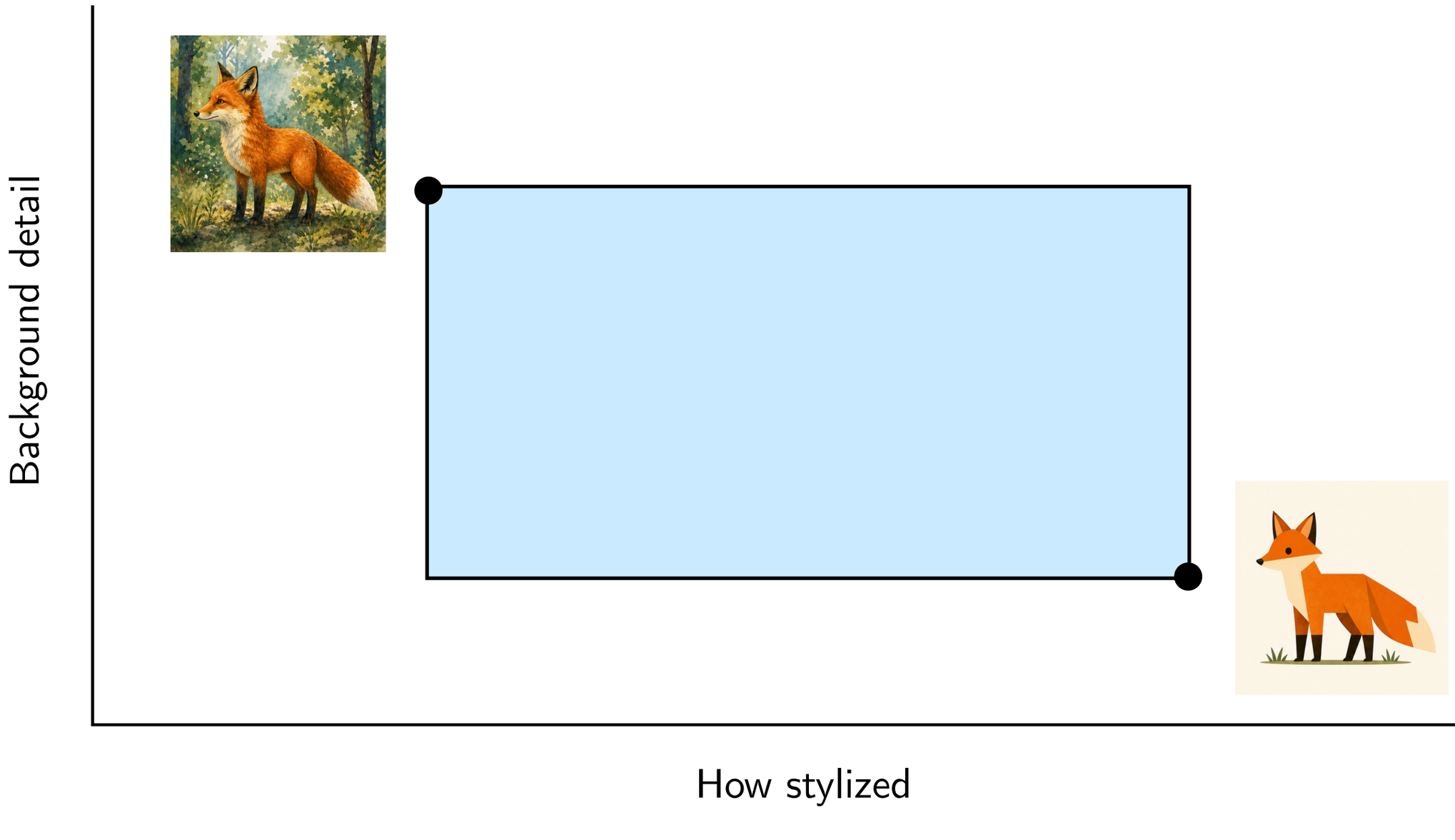
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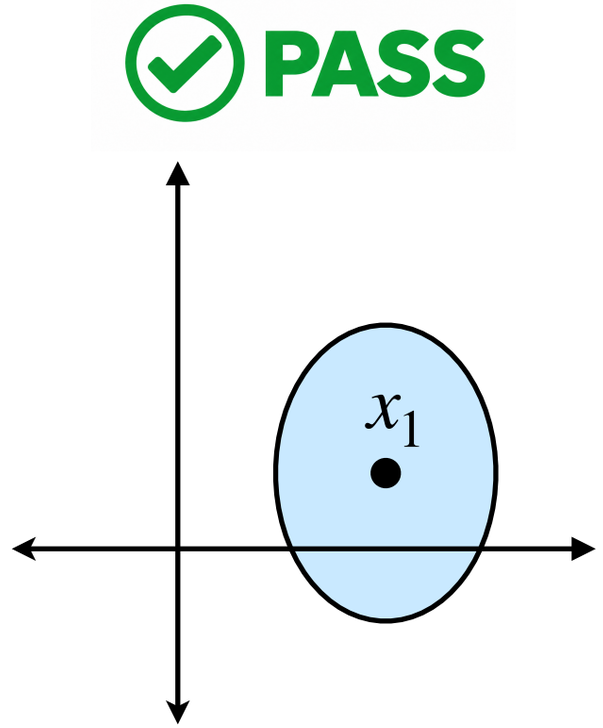


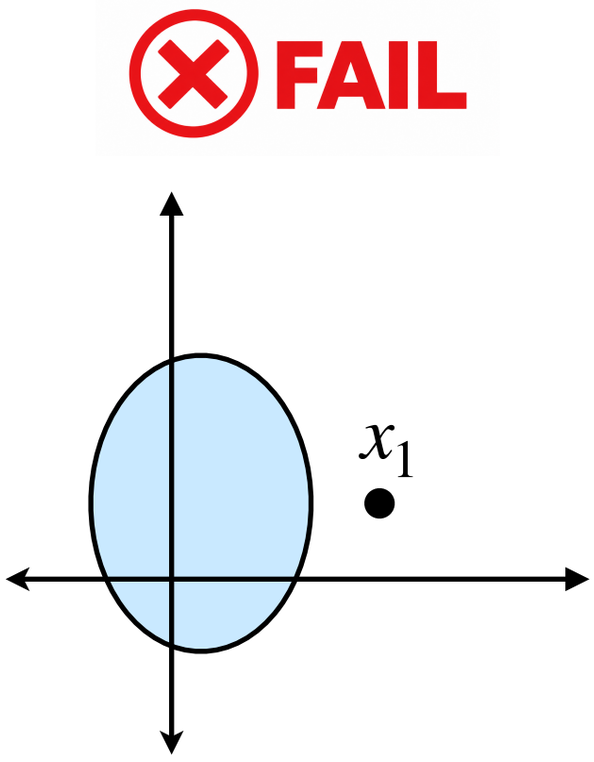
Admissible Generators

The *generator* g must satisfy three properties:

1. Preservation: $C \subseteq g(C)$

   All works in the training corpus are generable.





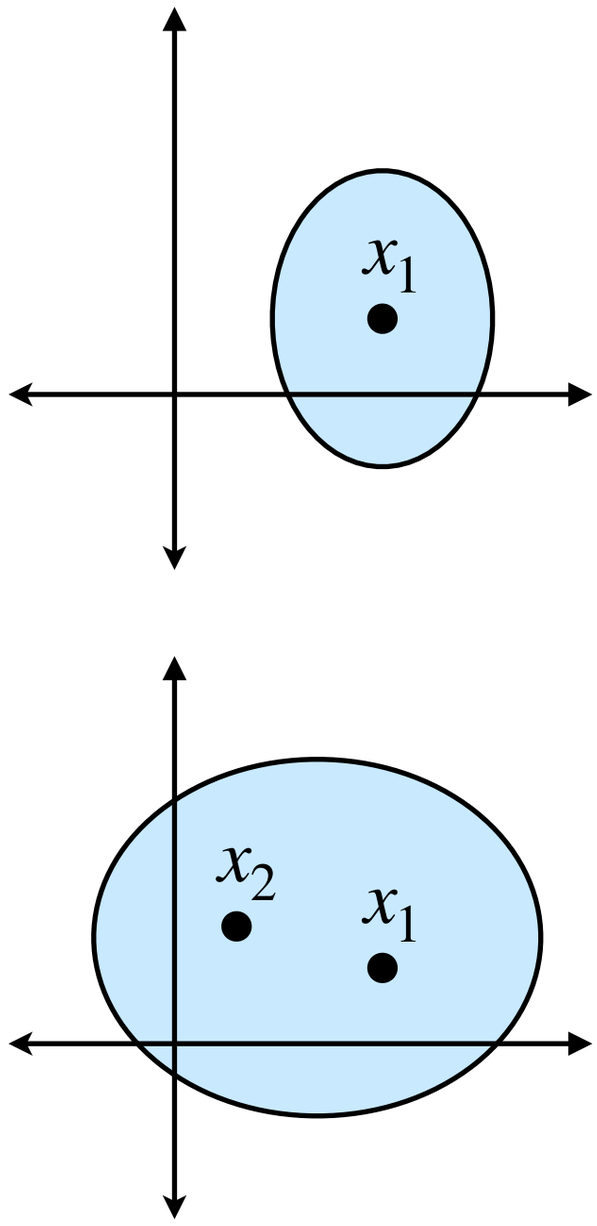
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Admissible Generators

FAIL

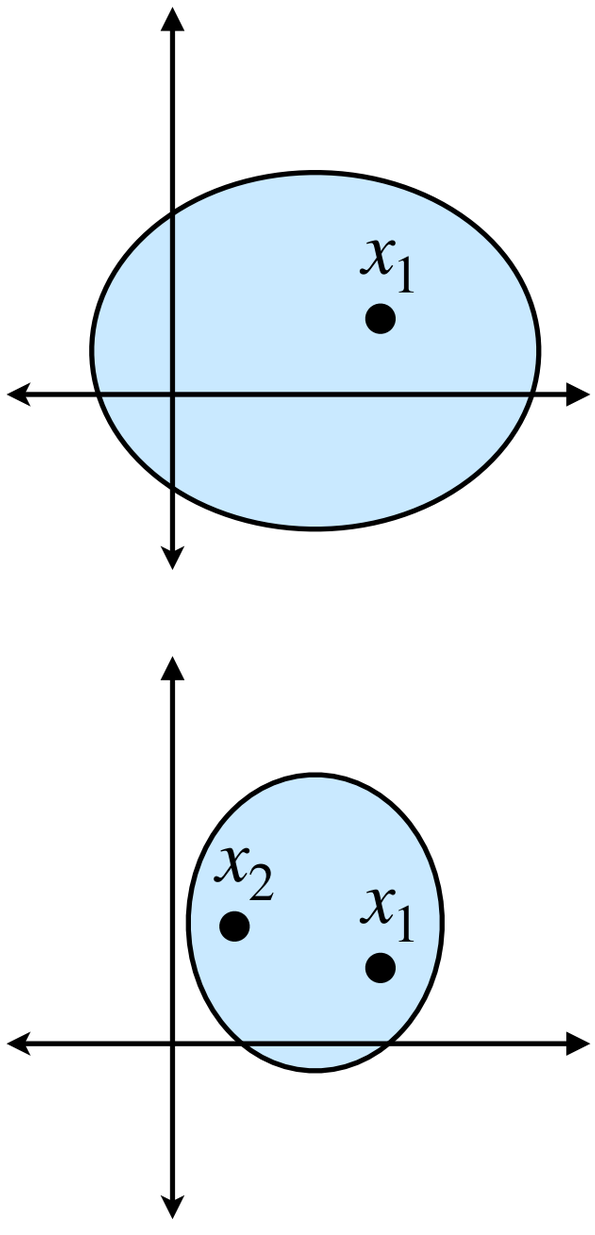
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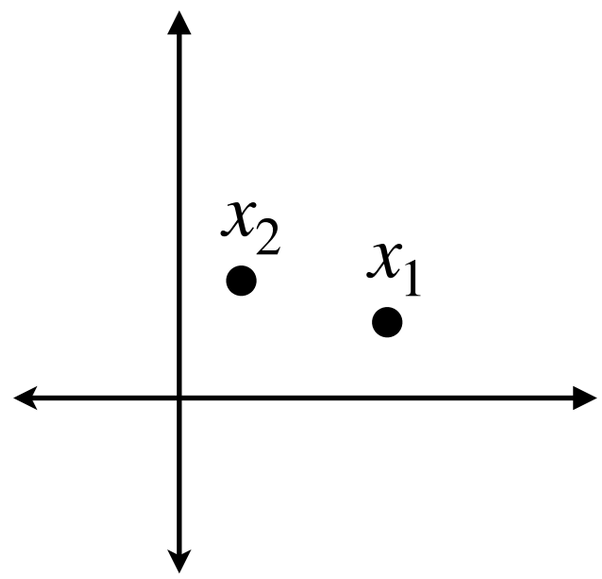
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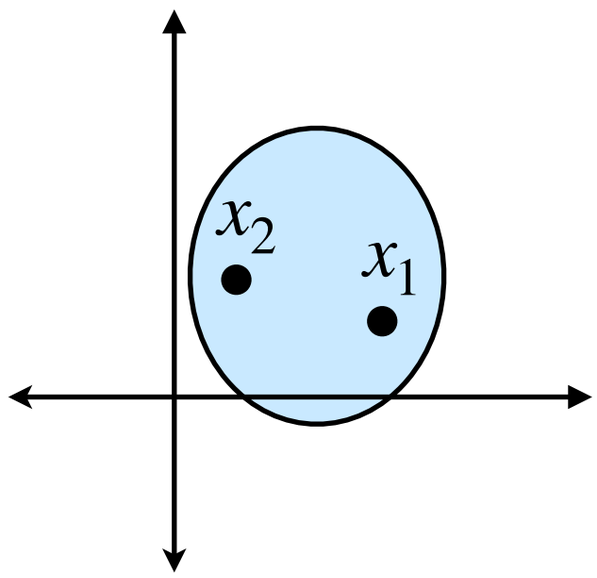
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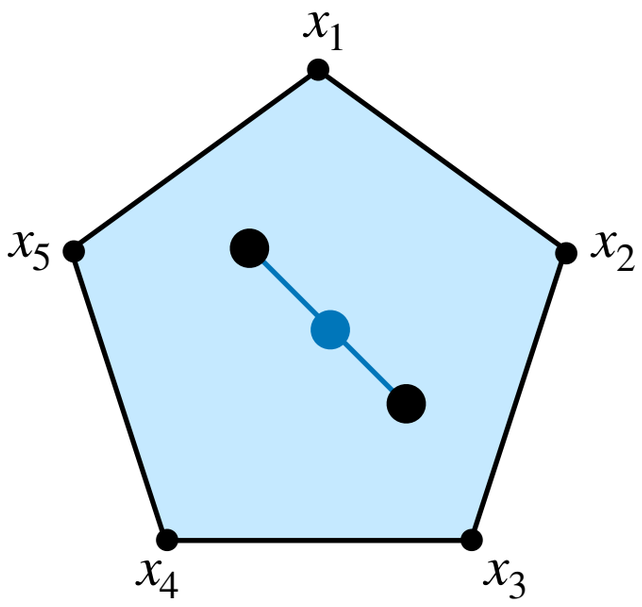


Convex-Valued Generators

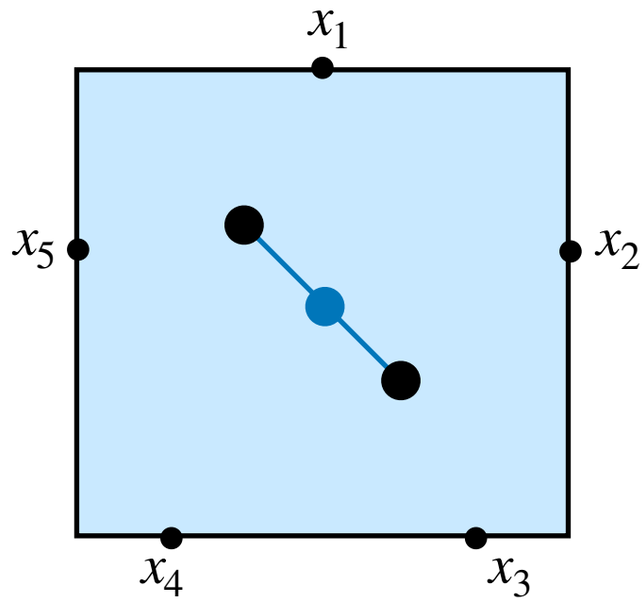
Definition: Say that a generator g is **convex-valued** if $g(C)$ is convex for every corpus C.

- We show that this is equivalent to the generator having the capability to take convex combinations of its own outputs.

Examples:



Convex hull generator



Box generator

Discussion

Is everything generable (e.g., monkeys typing Shakespeare)?

- Think of $g(C)$ as the creations that are generated with sufficiently high probability

What if you smuggle the desired creation in as the prompt?

- Restrict attention to "reasonable" prompts

Applying the test requires specifying a threshold for likelihood of generation and a criterion for admissible prompts.

Today, study the implications of the test once those choices are fixed.

Discussion

What does the analyst need to know to apply our criterion?

- With access to the AI generator, it isn't necessary to know exactly what g or the feature space are
- For some of our results, we will rely on knowledge of qualitative properties of the AI generator (e.g., convex-valuedness)

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Could this model describe human generation?

- In principle, yes! (e.g., $g_{Jay}(g_{Annie}(C))$ describes everything Jay can generate given Annie's ideas given the initial corpus C)
- But our subsequent definition will counterfactually vary the training corpus C, which can be done with AI generators and is not feasible with humans

Part II: Our definition of “derivative” generation

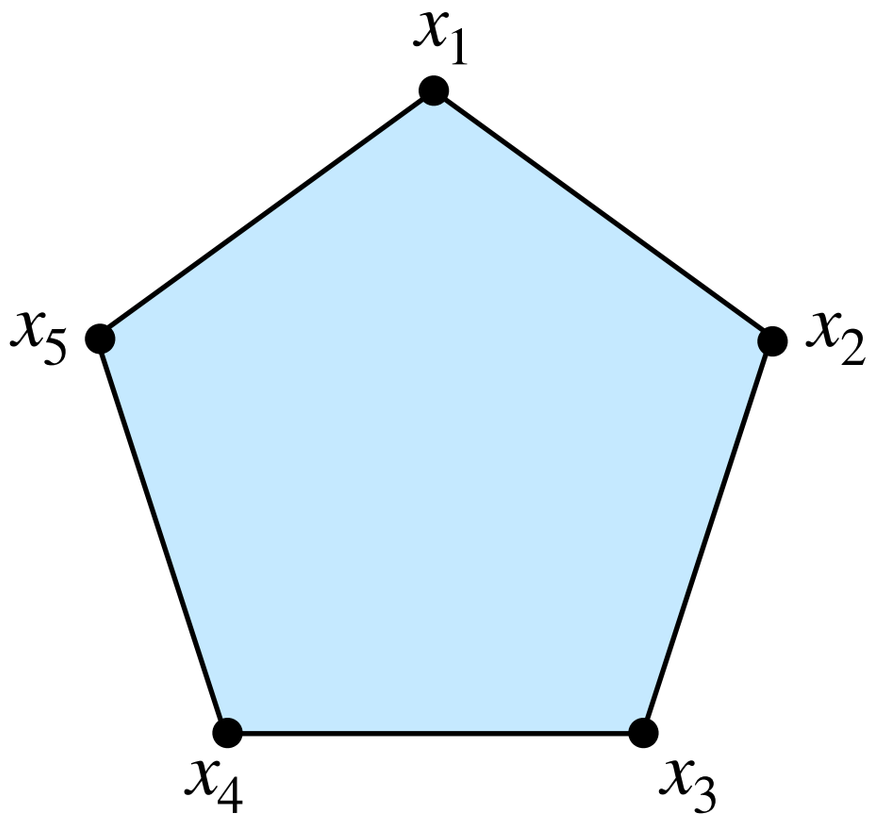
Permissible Generation

Definition: Fix any generator g, corpus C, and work $x \in C$. The x-**permissible set** is

$$p_g(x, C) := g\left(C \backslash \{x\}\right)$$

i.e., all creations that could have been generated without x

Convex hull generator:



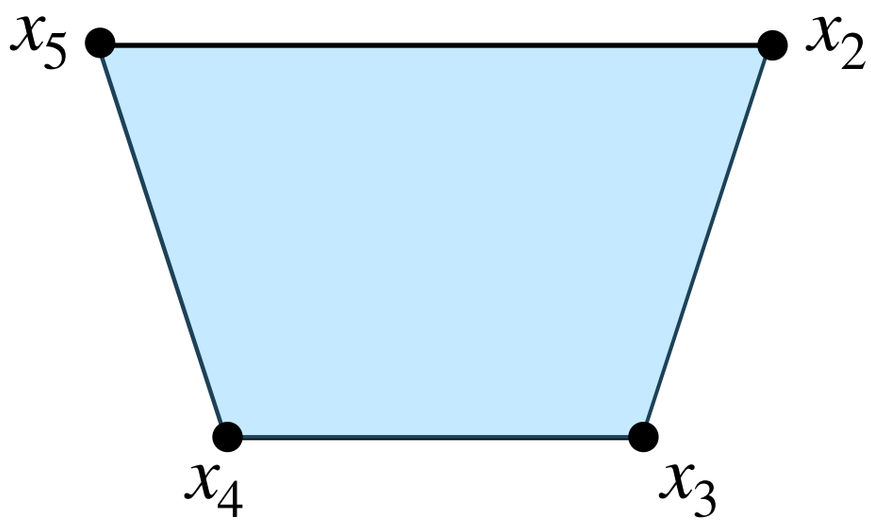
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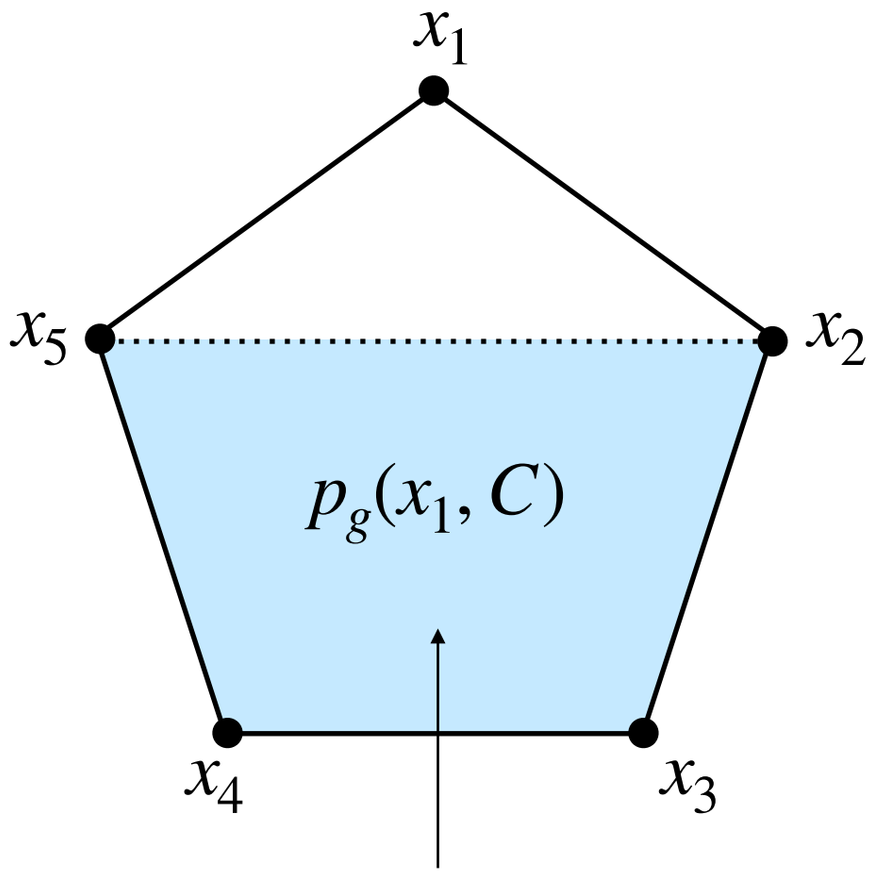
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Convex hull generator:



x_1-permissible set

(generable creations that do not require x_1)

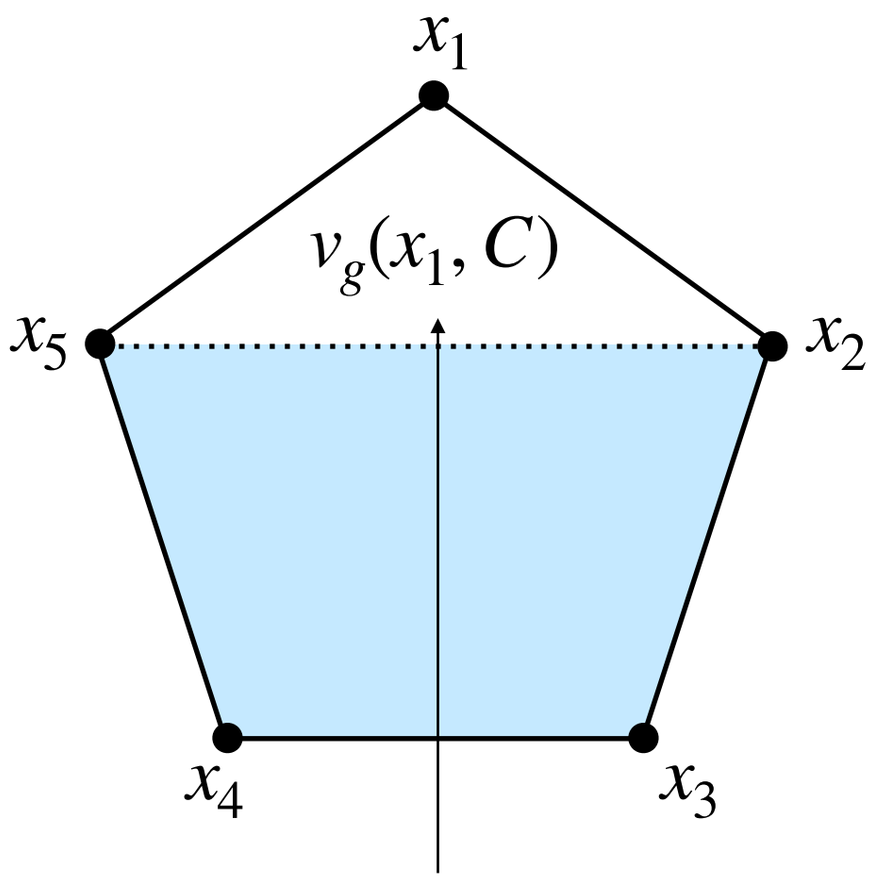
Derivative Generation

Definition: Fix any generator g, corpus C, and work $x \in C$. The x-**violation set** is

$$v_g(x, C) := g(C) \backslash p_g(x, C)$$

An output y **depends essentially** on x given C if $y \in v_g(x, C)$

Convex hull generator:



x_1-violation set

(creations that are generable only because of training on x_1)

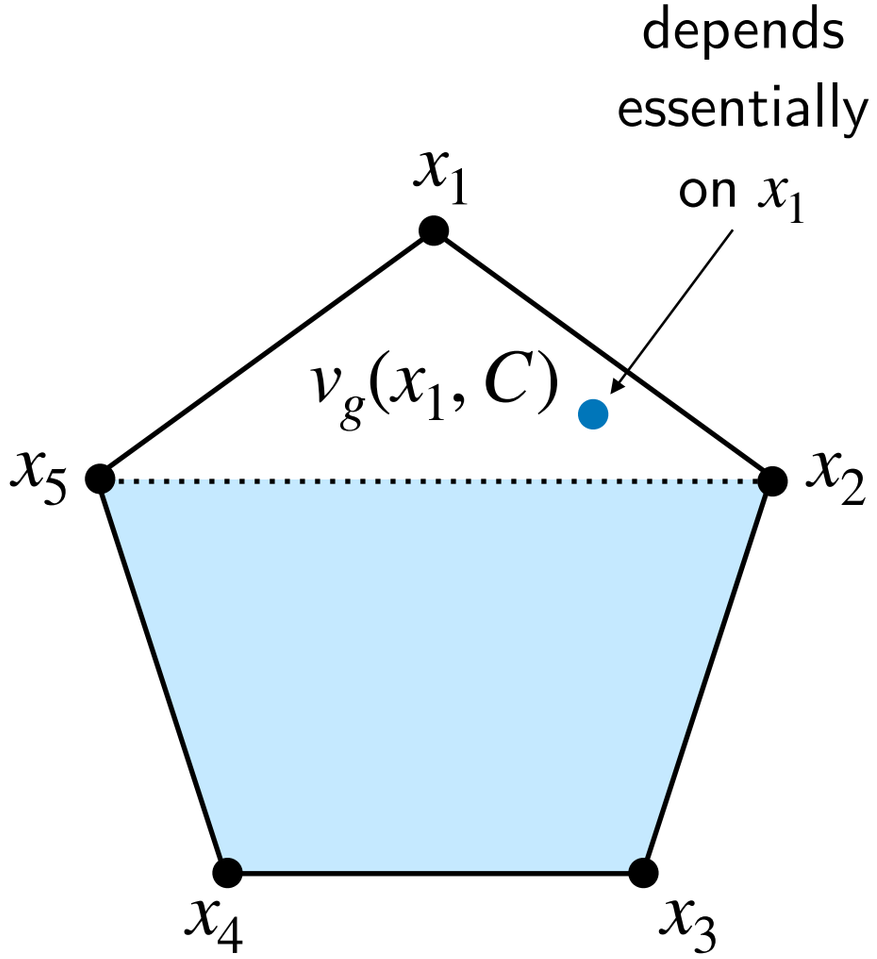
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Convex hull generator:



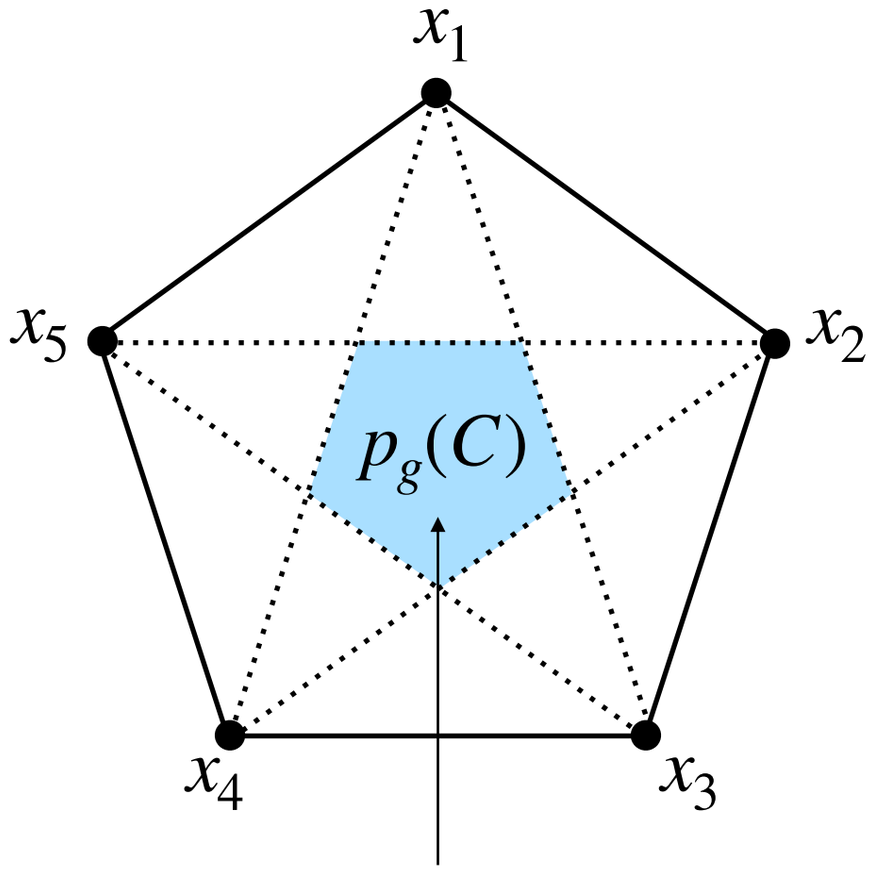
Permissible Generation

Definition: Fix any generator g and corpus C. The **permissible set** is

$$p_g(C) := \bigcap_{x \in C} p_g(x, C)$$

i.e., all creations that do not depend essentially on any $x \in C$

Convex hull generator:



permissible set

(generable creations that do not critically rely on any $x \in C$)

Discussion

Q: What are the consequences of depending essentially on an existing work?

- The predecessor may be entitled to **attribution**.
- The predecessor may be entitled to **financial compensation or rent**.
- The downstream work may be **restricted or prohibited**.

Our framework is agnostic about which.

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Q: How would permissibility be evaluated in practice?

- Brute-force retraining of an algorithm is not feasible
- Machine unlearning provides methods to approximate the role of specific training points (Bourtoule et al., 2019; Xiong et al., 2024)
- Can also view permissibility as an evidentiary question and compare outputs across independently trained algorithms

Restrictiveness

- Our criterion is not ranked in restrictiveness relative to the restrictions on training and outputs that we discussed
- Compared to a blanket ban on training on copyrighted works
  - We do not restrict use of copyrighted works, as long as the generation is in the permissible set **(less restrictive)**
  - We restrict generation even if the AI system was not trained on any copyrighted works **(more restrictive)**

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  - The violation set can generally be larger than the set of works that are "substantially similar" to an original **(more restrictive)**

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- We view our criterion as a complement rather than a substitute for the existing criteria, could be that all should be simultaneously enforced

Part III:

Properties of Permissible Generation

Are **any** AI generations permissible?

- An output is permissible only if it passes a demanding test: it must remain generable after removing any individual work
- One concern might be that nothing will pass

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- One concern might be that nothing will pass

Example: The generator is the convex hull generator, and the corpus consists of two works:



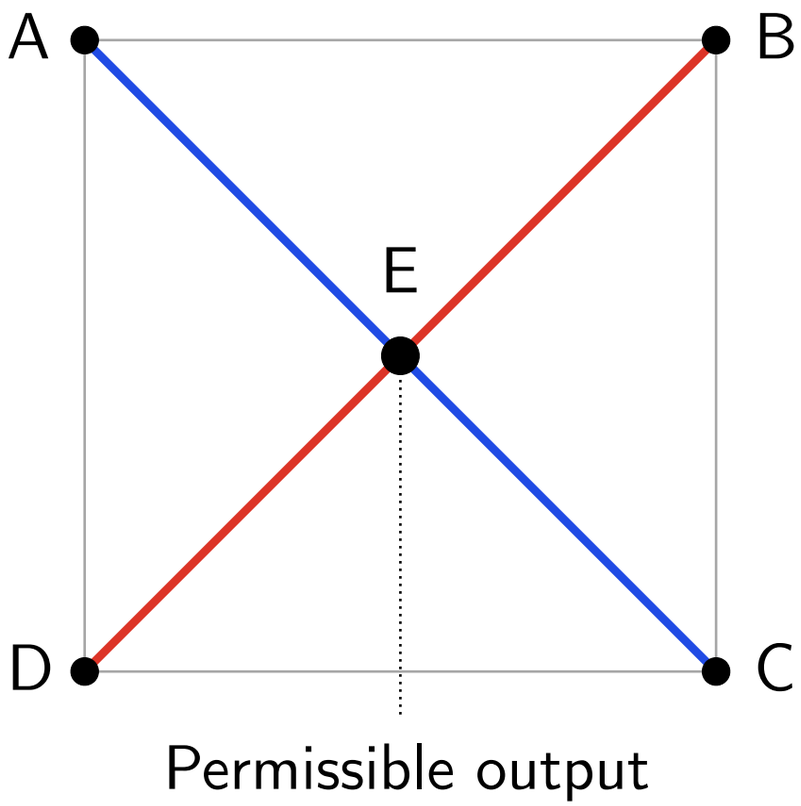
The x_1-permissible set is $\{x_2\}$, and the x_2-permissible set is $\{x_1\}$, so their intersection is empty.

No generations are permissible.

Are **any** AI generations permissible?

Result: With a convex-valued generator in d dimensions, any corpus of at least $d + 2$ works has a permissible output.

- Generalizes to arbitrary generators where $d + 2$ is replaced by the generator's Radon number



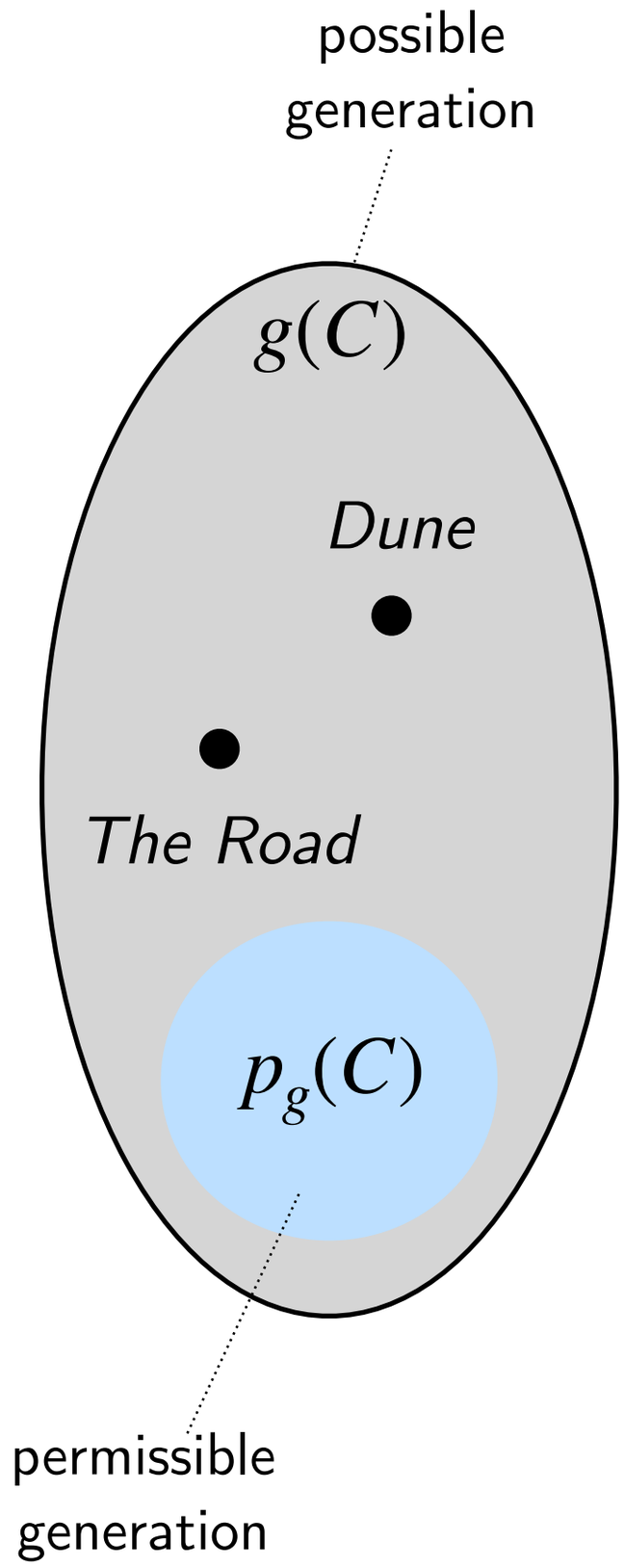
Two routes to the same output E:

- Combine A and C
- Combine B and D

Generation of E remains possible no matter which initial work is removed.

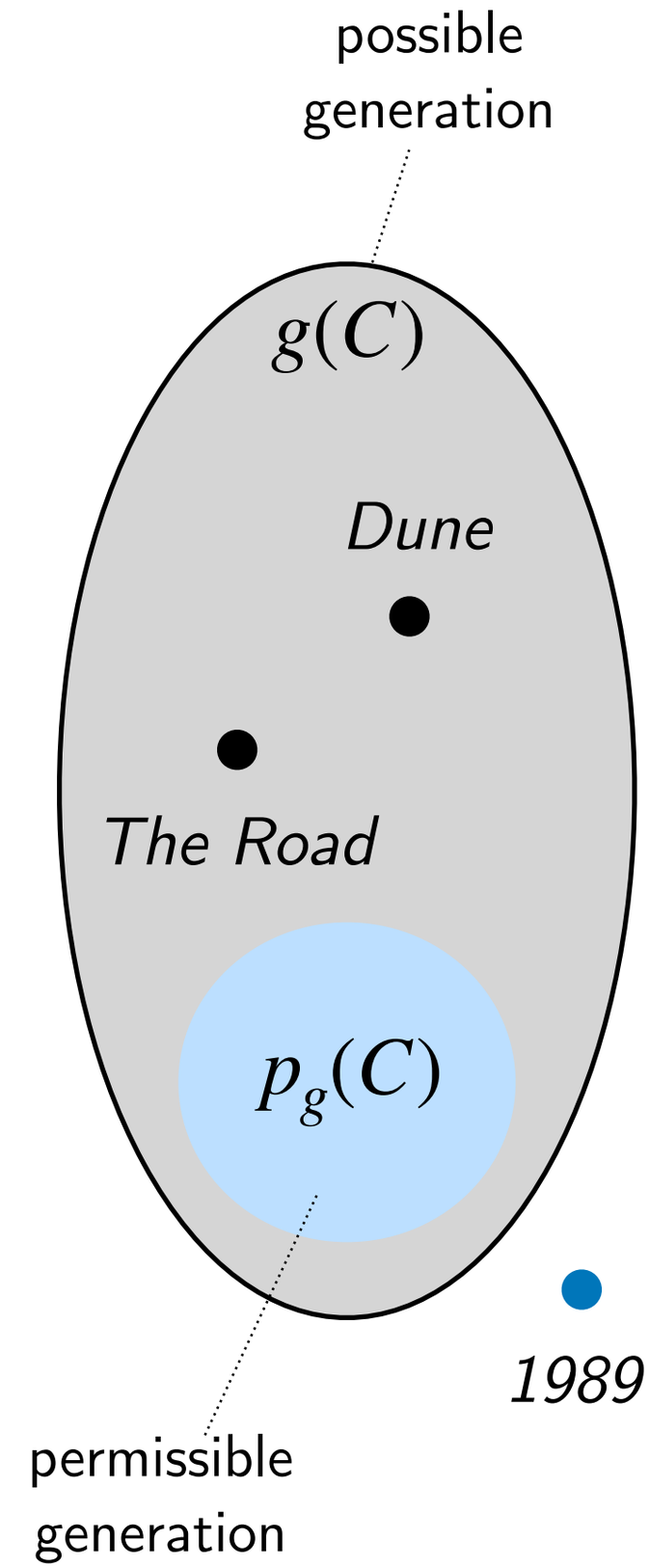
What happens as we train on more works?

- Suppose we add a new work to the training corpus of the AI system.
- Will this shrink or expand the set of permitted generations?



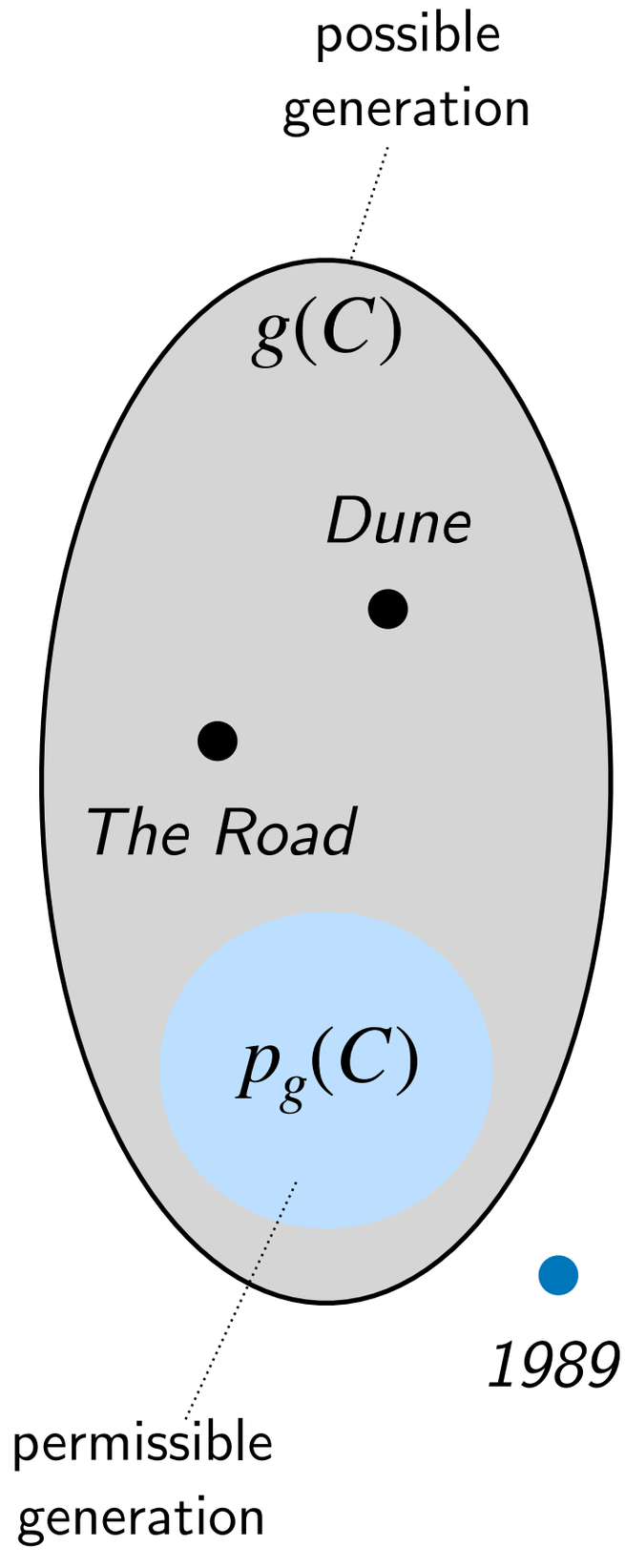
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What happens as we train on more works?

- Suppose we add a new work to the training corpus of the AI system.
- Will this shrink or expand the set of permitted generations?
- Why it might expand: the AI system is physically capable of generating more works (by assumption that g is monotone)
- Why it might shrink: This new work's rights also have to be respected, hence an additional restriction on generation.
- The former wins out—**the set of permissible creations can only grow**.



What happens as we train on more works?

Result: The permissible set is **monotone**: $C \subseteq D \implies p_g(C) \subseteq p_g(D)$

i.e., adding new works to the corpus weakly expands the set of permissible generations.

Intuition:

- The new work cannot claim it was essential to something that was already permissible before it arrived.
- But it may provide an alternative way to generating something that previously depended on another work.

Part IV:

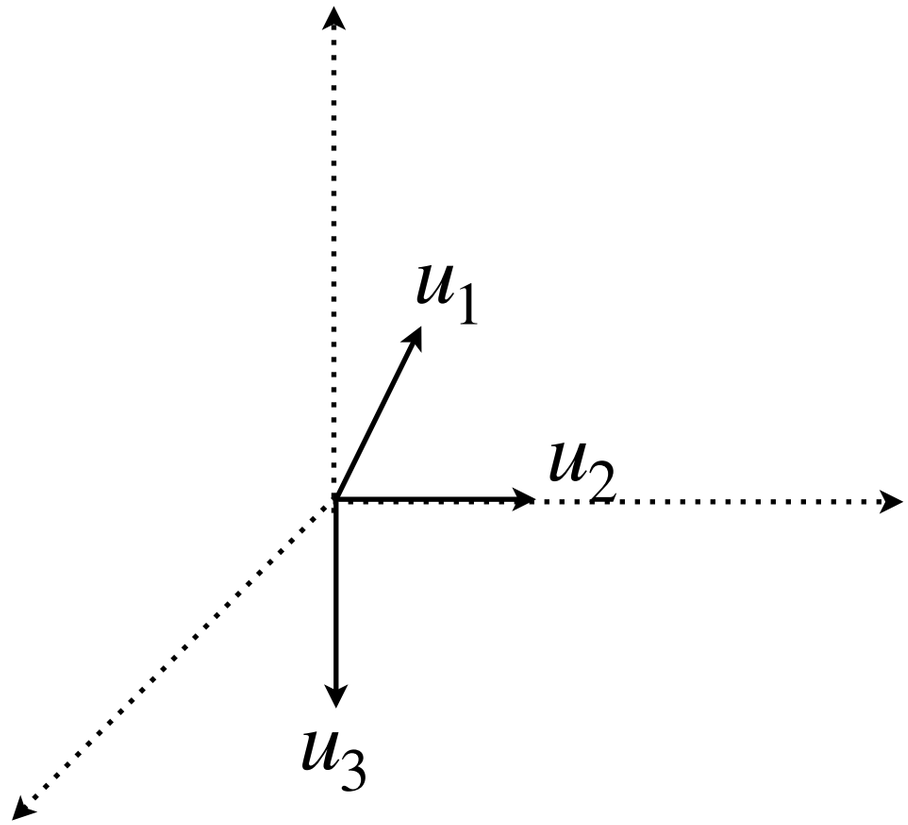
Permissible Generation given Large Training Corpora

Does scale make individual works replaceable?

- We just saw that adding more works to the AI's training corpus weakly expands the set of permissible generation
- Does this mean that as the AI is permitted to train on more and more works, eventually most individual contributions will be dispensable?
  - In this case, our criterion would cease to place meaningful restrictions.
- To answer this, we:
  - model the creative process by which the training corpus C is generated (i.e., how individual creators create new works)
  - study how permissible generation evolves as this corpus grows large

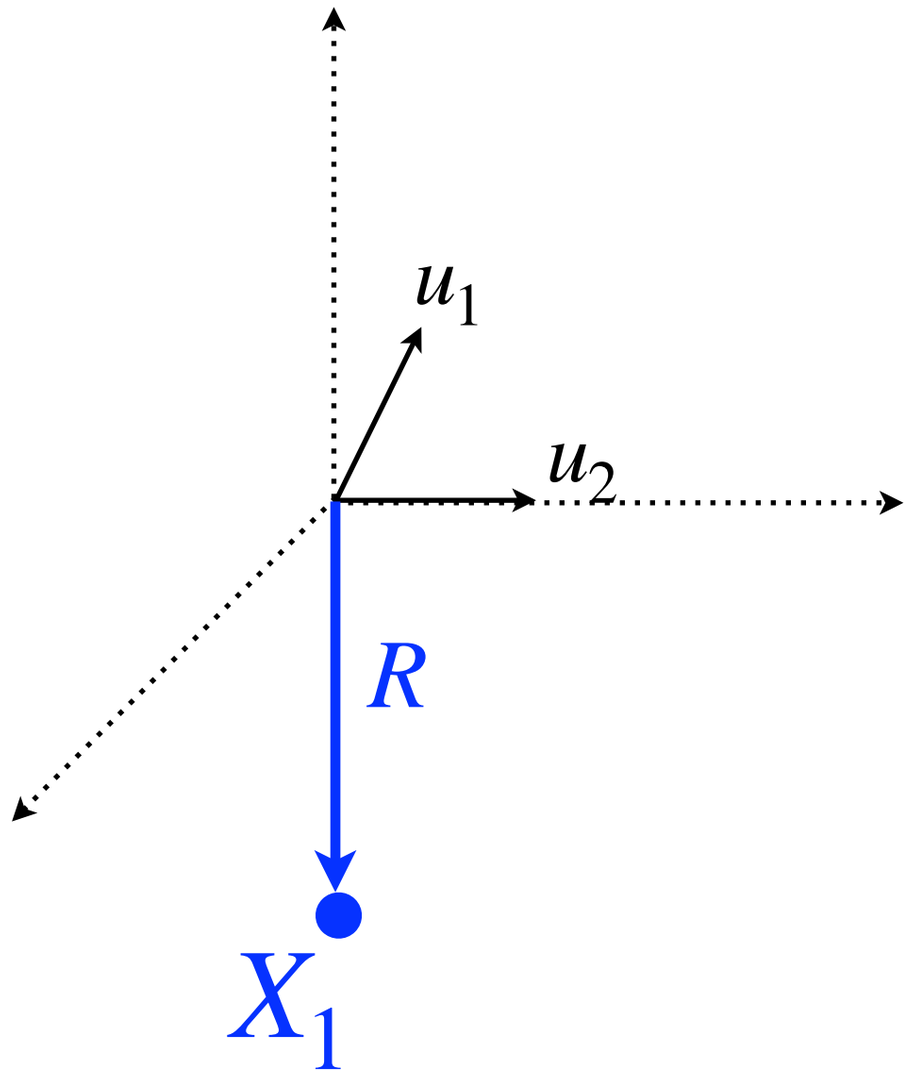
The Creative Process (Informal)

- There are d creative **directions** (unit vectors) spanning the creative space $\mathbb{R}^d$
- At each date n, a creator:
  - Chooses a creative direction
  - Innovates in that direction by a **random magnitude**
- The date n corpus consists of all works generated up to that date



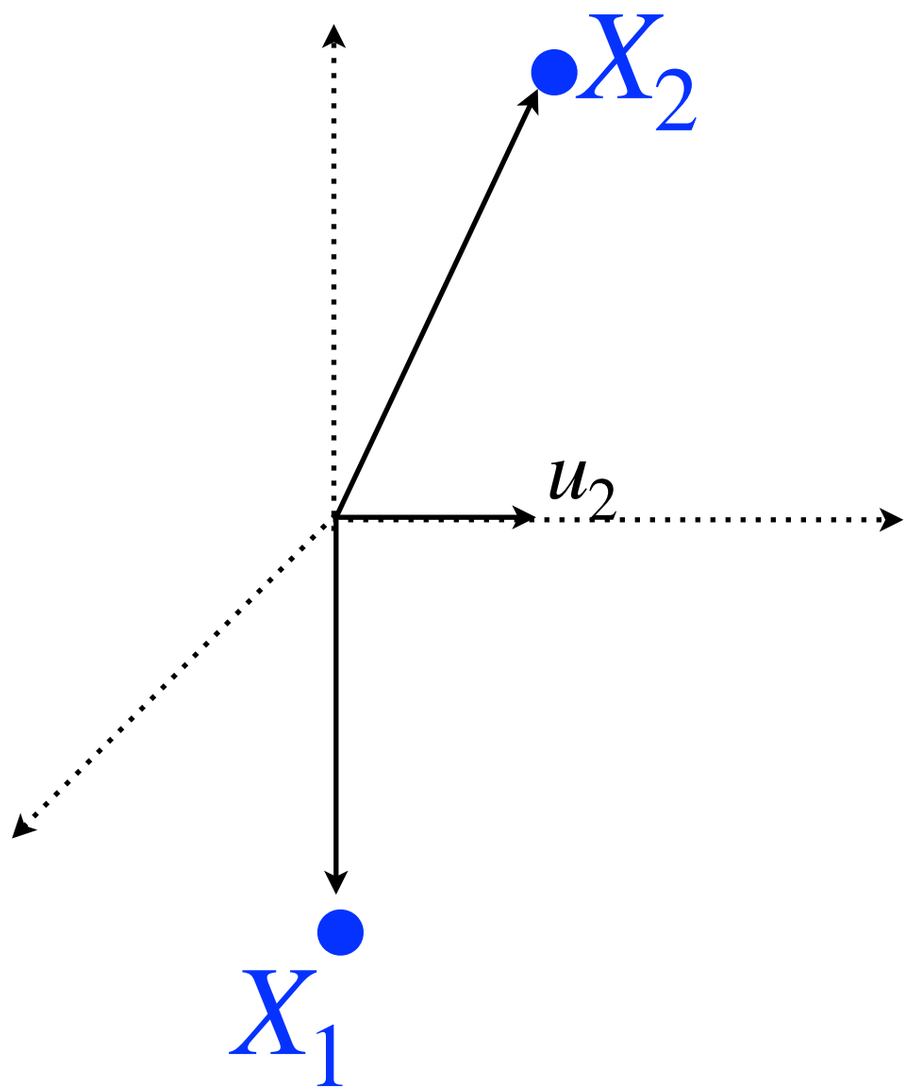
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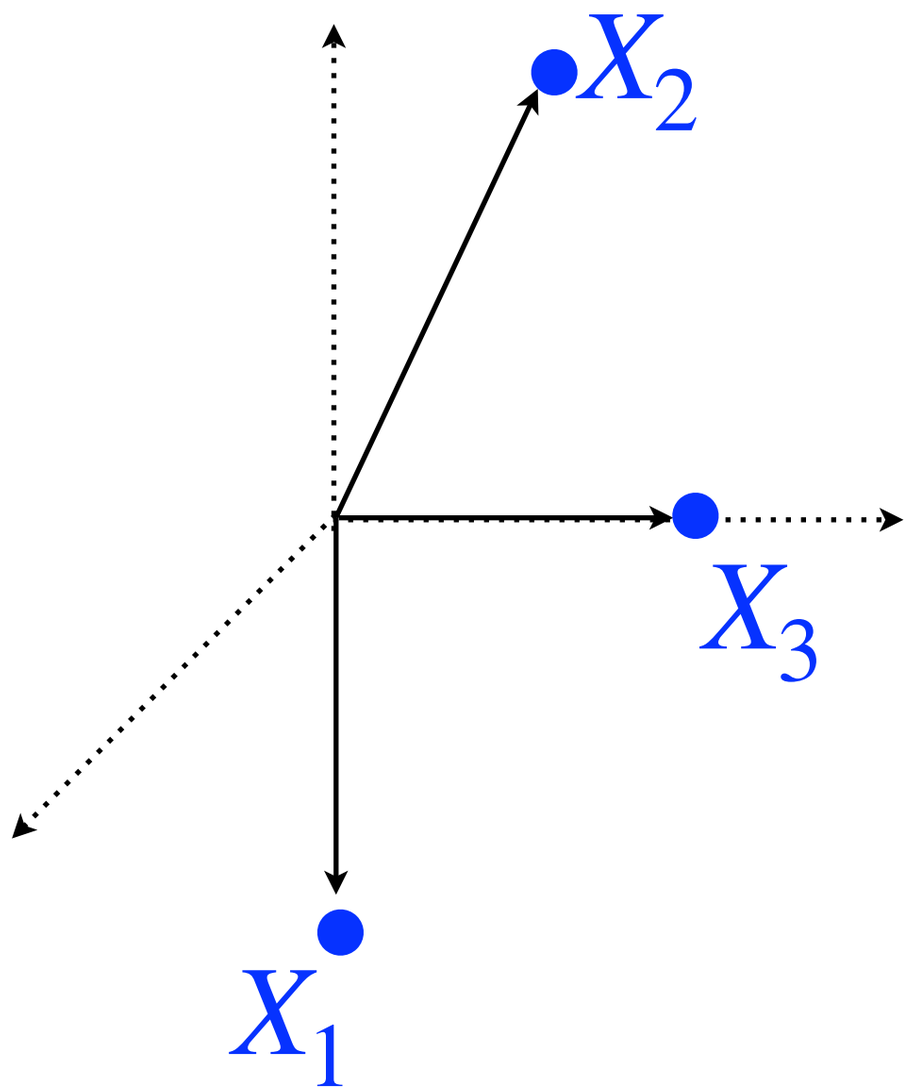
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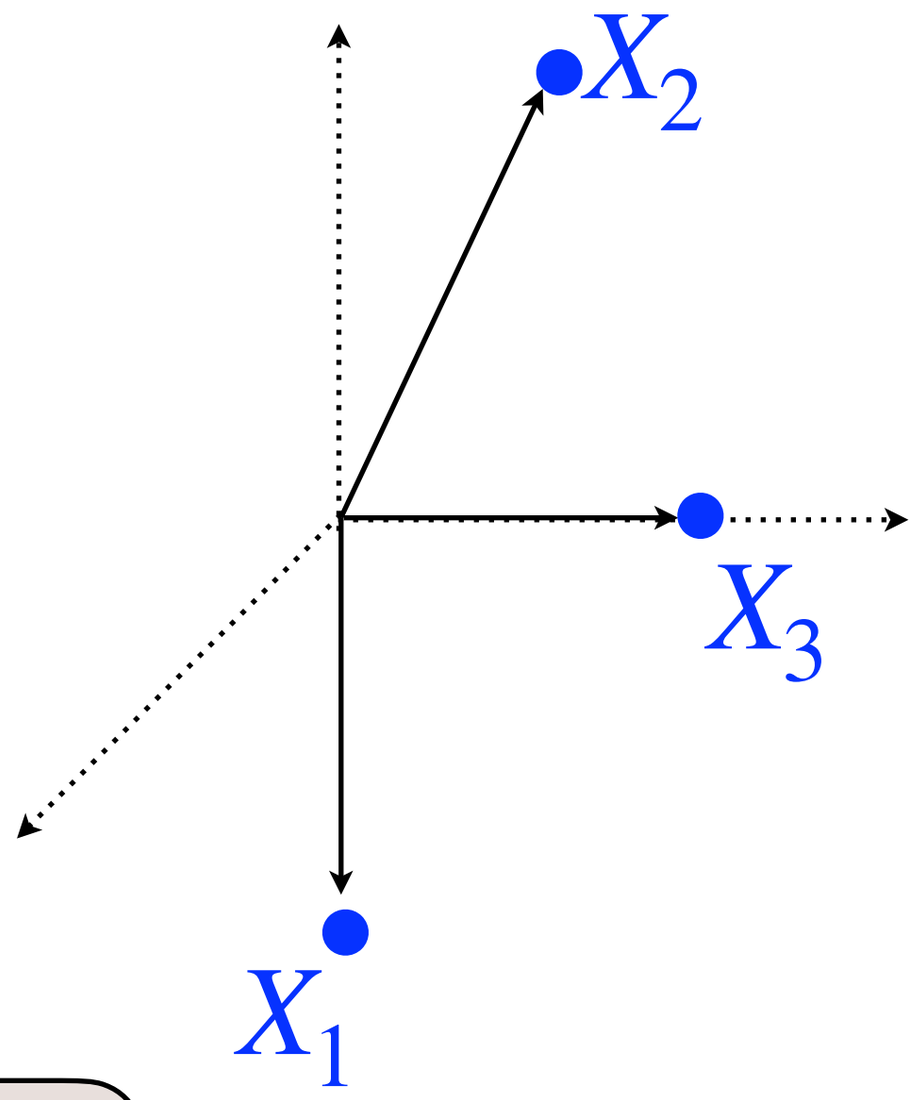
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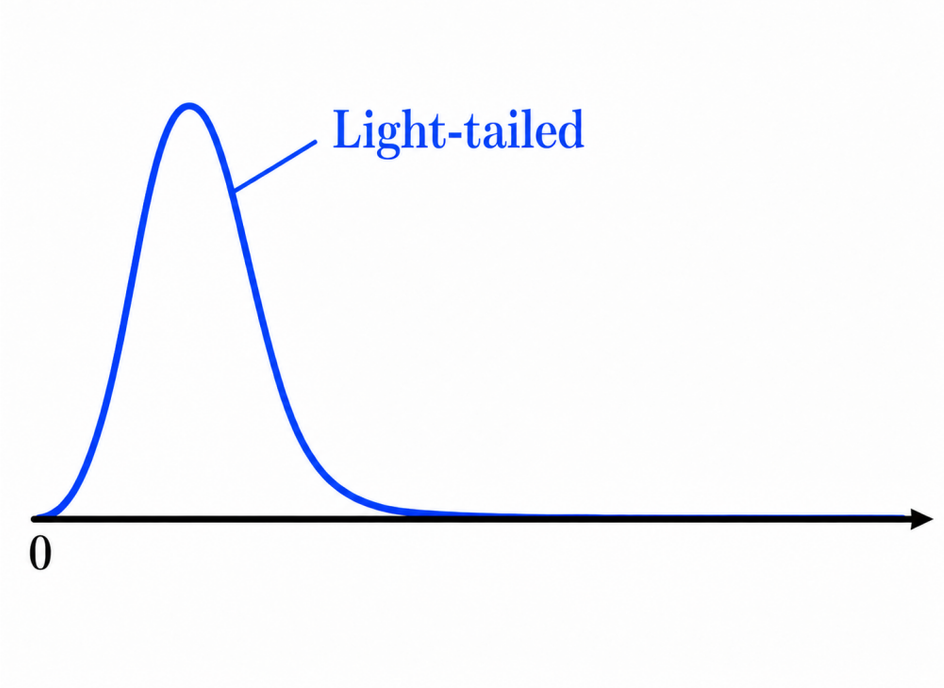


The distribution of these innovations is crucial.
We consider two cases: **light-** and **heavy-tailed**

Light-Tailed Distribution (e.g., Gaussian)

Definition: A distribution F on $\mathbb{R}_+$ has **light tails** if
$1 - F(ct) \leq (1 - F(t))^c$ for all $c \geq 1$ and $t \geq 0$ (e.g., truncated Gaussian)

- Key property: very large departures become increasingly less likely.
- Example: adult height

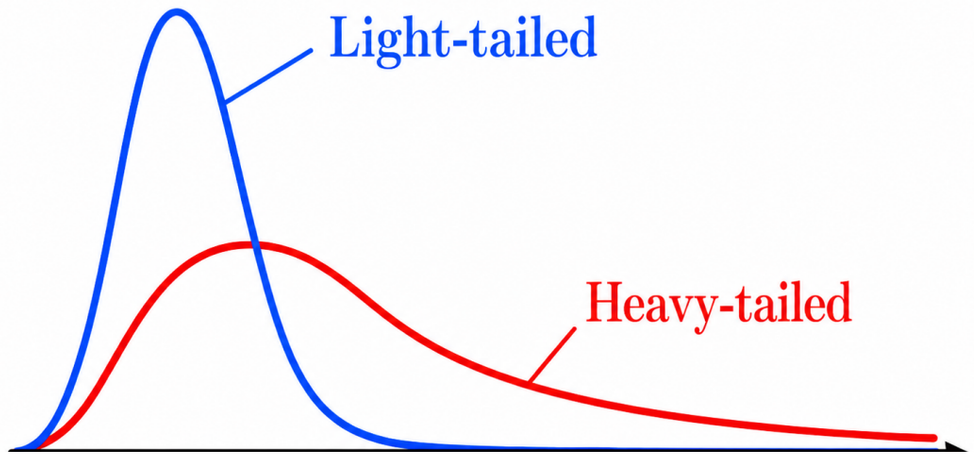


Possible creative process: music within an established genre.

Light-Tailed Distribution (e.g., Gaussian)

Definition: A distribution F on $\mathbb{R}_+$ has **heavy tails** if there is a constant $\alpha > 0$ such that $1 - F(ct) \geq c^{-\alpha}(1 - F(t))$ for all $c \geq 1$ and $t \geq 0$ (e.g., Pareto)

- Key property: a few observations can dwarf most others.
- Example: wealth, citations

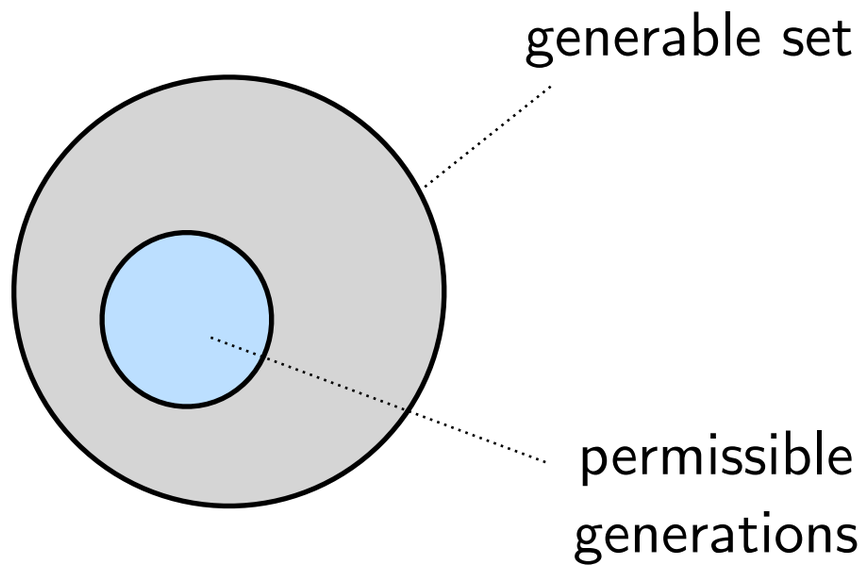


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Main Result

Theorem (informal): Fix a convex-valued generator g.

1. With light tailed innovation, the permissible share of the generable space **approaches one**.

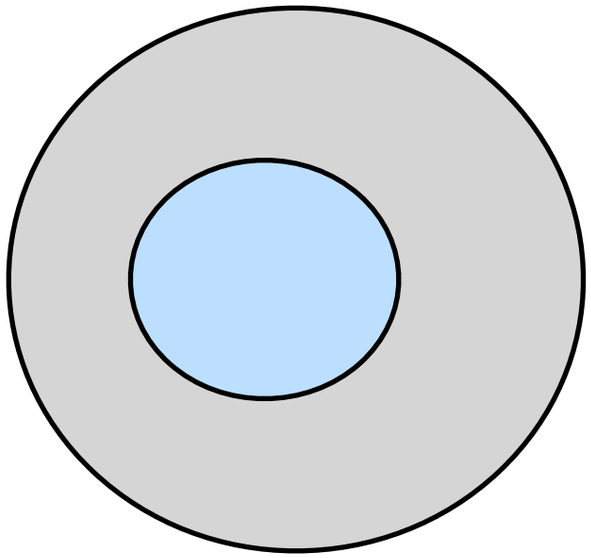


Share measured by volume. Conclusions hold with probability 1, assuming each creative direction is sampled infinitely often.

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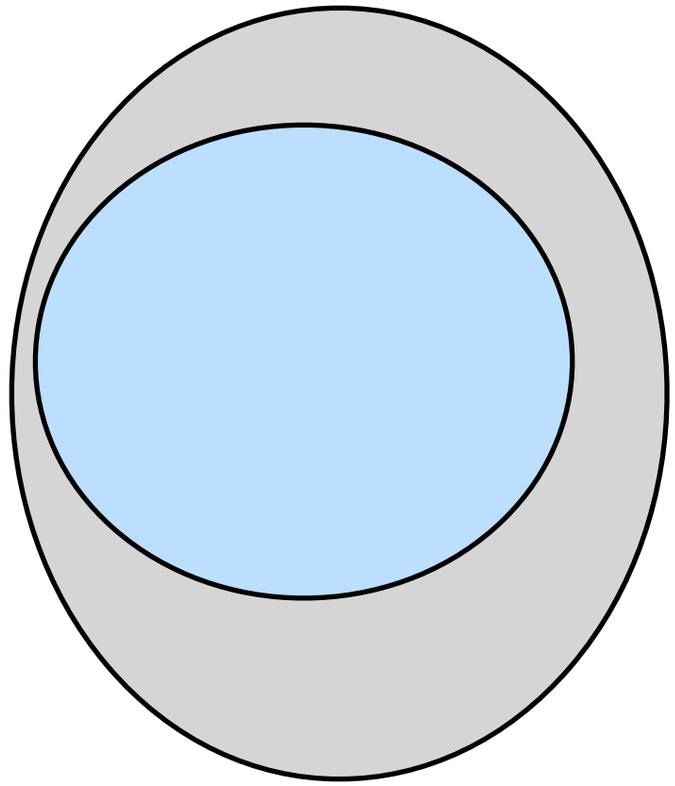


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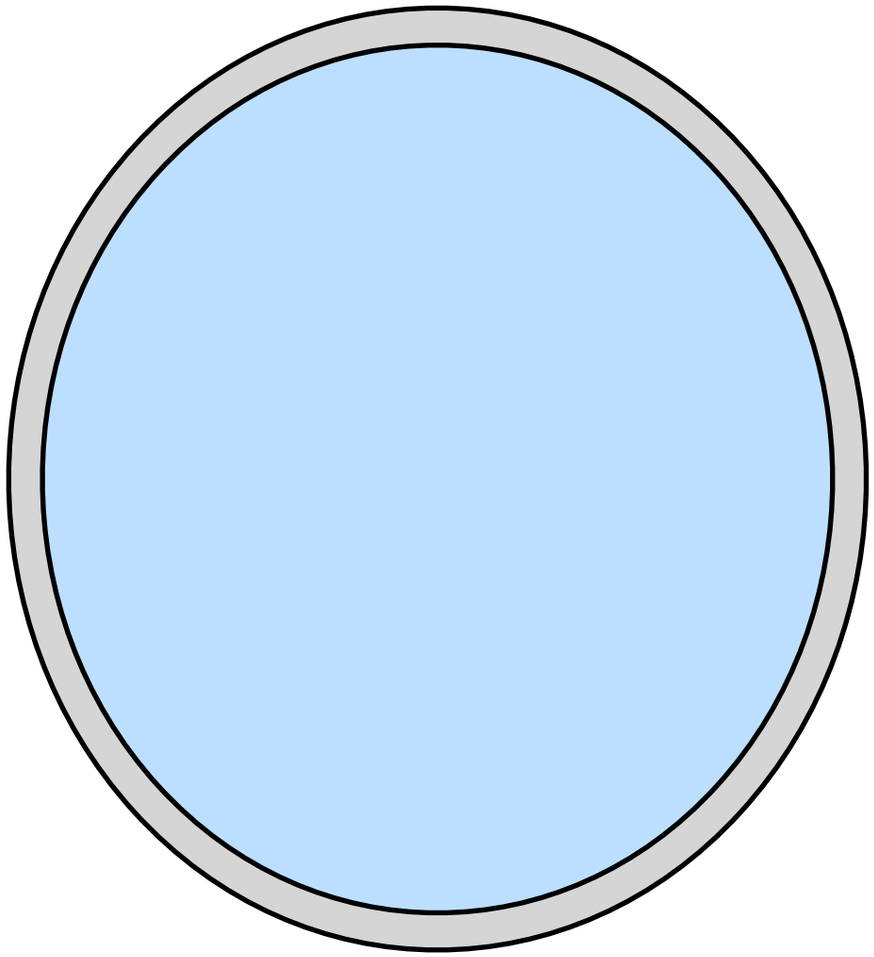


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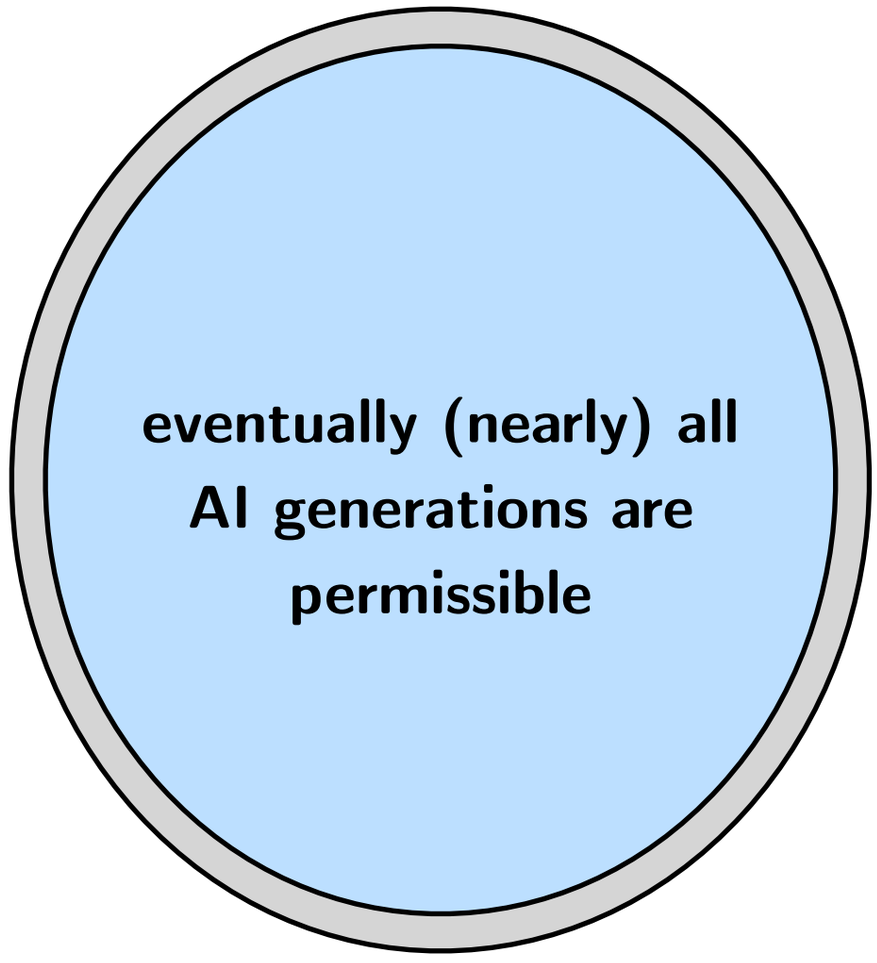


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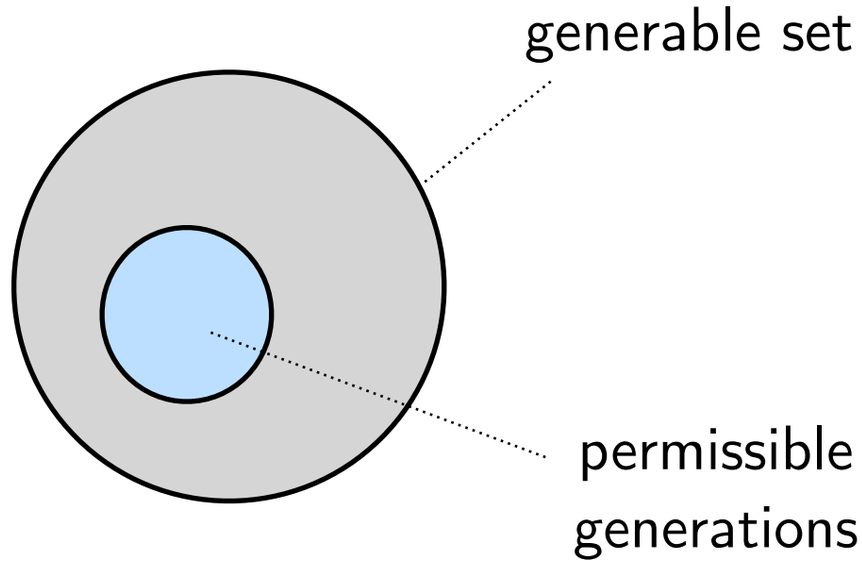


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Theorem (informal): Fix a convex-valued generator g.

1. With light tailed innovation, the permissible share of the generable space **approaches one**.
2. With heavy tailed innovation, this share **repeatedly falls arbitrarily close to zero**, even as the corpus grows arbitrarily large.

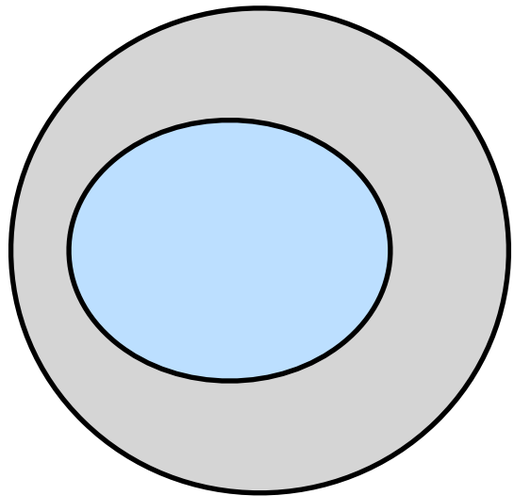


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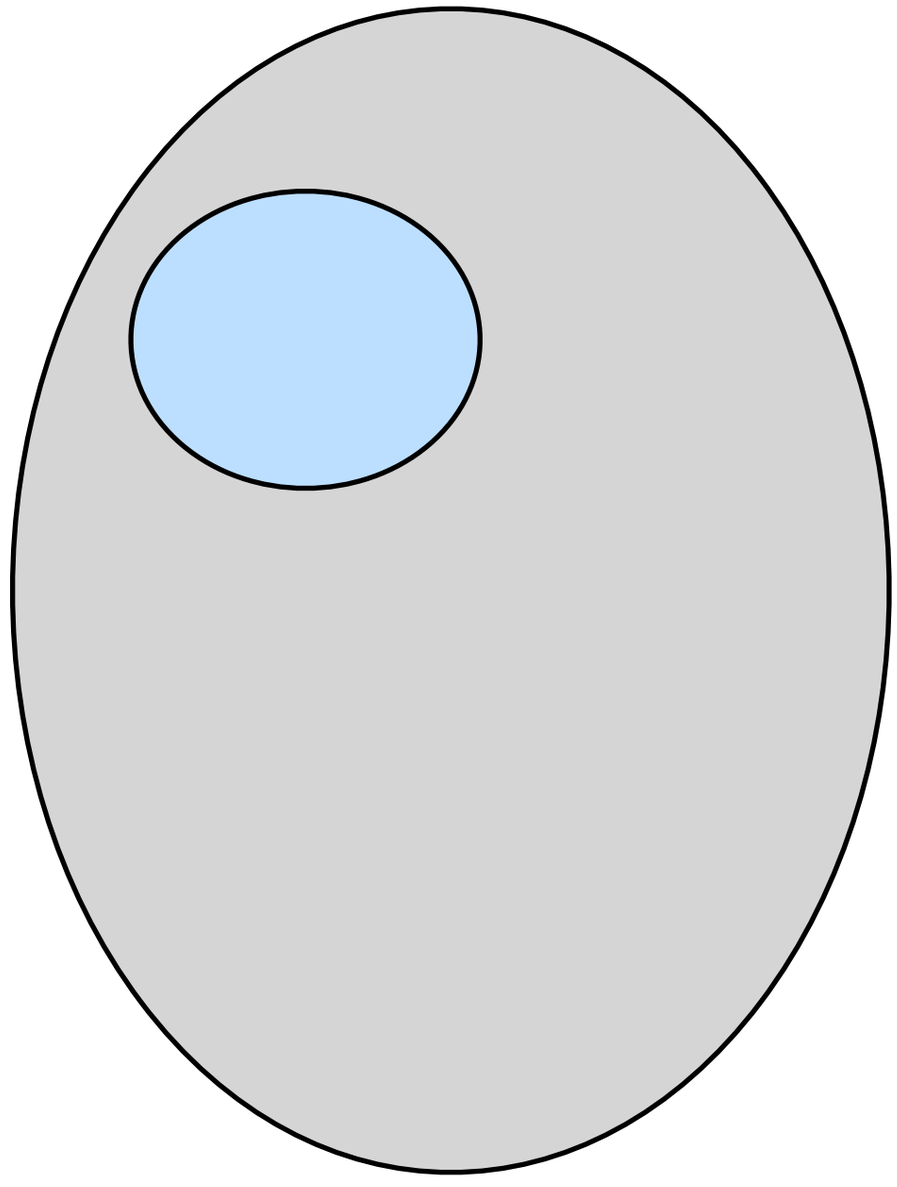
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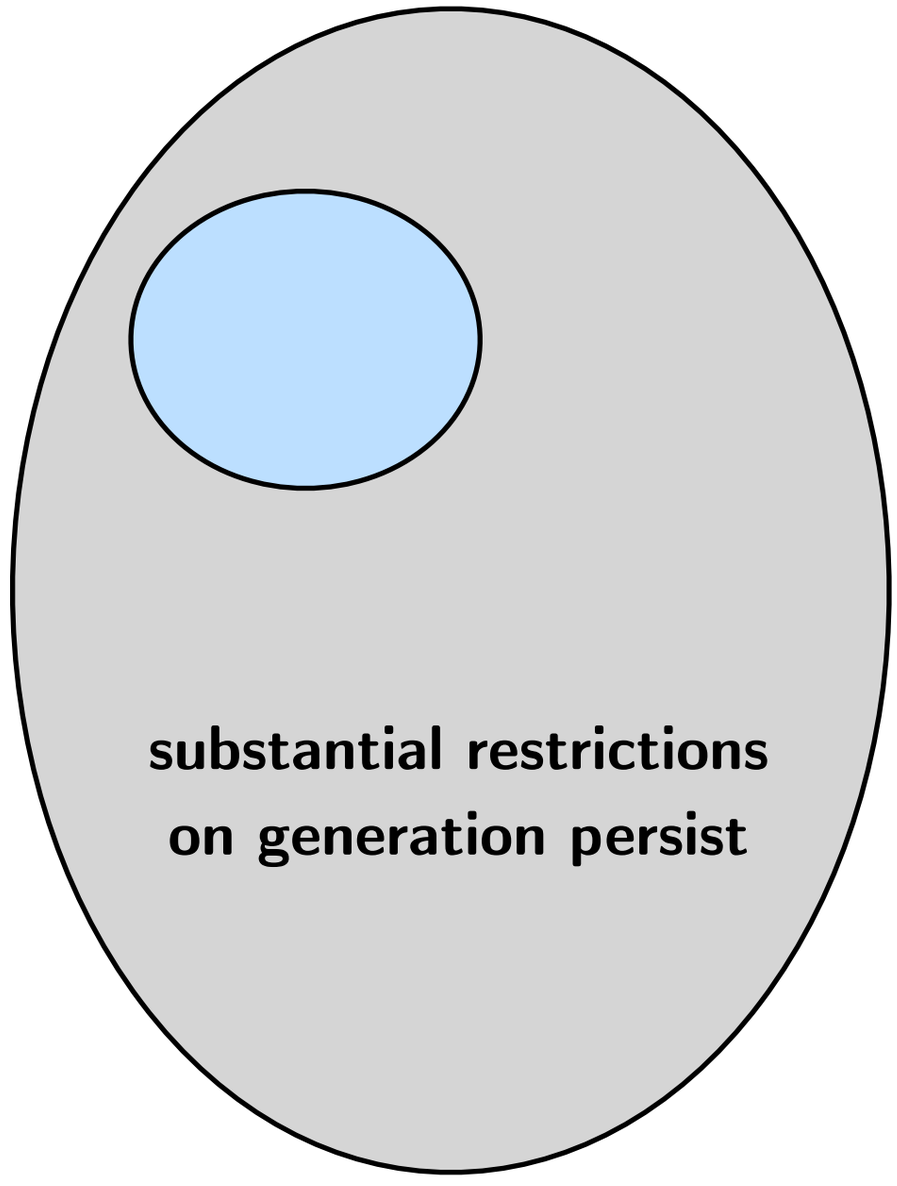
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Intuition

- With light tails, even the leading works eventually have **close alternatives.**
    - Removing any one work makes only a small proportional difference.
- With heavy tails, **exceptional works keep arriving** that expand what the corpus can generate.
    - Much of that expanded space can depend on a single work.

Takeaways

- In creative markets where innovation is light-tailed, individual creators are eventually marginal
- If there are rare but significant breakthroughs,
  - individual creators remain pivotal
    - our criterion strongly binds even with arbitrarily large corpora
- We've taken the regime as given, but could also consider the question of which we would expect to emerge endogenously
- e.g., incentives for creators might push towards a heavy-tailed regime (especially if creative protections follow our proposal)

Part V:

Creative Protection for Bodies of Works

Generalization to Protected Bodies of Work

So far, we have tested dependence on one work at a time.

Our criterion is easily generalized to creative protection for a collection of works, e.g.,

- A creator's full catalogue



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- A creator's full catalogue
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New primitive: a family of sets of works $\{A_1, \ldots, A_K\}$ where each A_i is a **protected group** (e.g., all Disney characters)

The permissible set is the intersection of the A_i-permissible sets (rather than the x_i-permissible sets)

Can a Catalogue be Indispensable?

Main result: violations are superadditive

Intuition: If two works are substitutable, then neither will be critical for generation of any output. But it can be that the **pair** is indispensable.

What is removed?	Can the output still be generated?
Work A	Yes, using Work B
Work B	Yes, using Work A
Both works	**No**

Implication: if a firm that owns A_1 merges with a firm that owns A_2, the permissible set of generations becomes (weakly) smaller

Conclusion and Open Questions

- Our paper proposes a new definition for when a generated work is "derivative" of an original
- Directions for extension:
  - Endogenizing the generator and the distribution of creative works.
  - Suppose creators are paid as a function of how much their works expand the generable set. What works would be created? What coalitions optimally form?
  - Cardinal measures of influence

Thank You

Can combining permissible works yield a violation?

- Imagine collecting every output that passes our test and using that collection as training material for another round of creation
- Could combining those works produce something that depends essentially on one of the original works (hence, a violation)?

Result: The permissible set is **self-stable**: $g(p_g(C)) = p_g(C)$

i.e., generating from all permissible works yields only those works